

KROZENDA

MULTI-VENDOR B2B & B2C MARKETPLACE PLATFORM

Product Requirements Document

The complete functional and technical specification for the Krozenda platform — every module, every role, every flow, every business rule, and every acceptance criterion for the contracted Phase 1 delivery.

Covers three simultaneous business models — Direct Dropshipping, Marketplace, and Own Stock — across one catalog, one cart, one checkout, and one settlement engine, delivered as a responsive web platform plus four native app binaries.

Document	Product Requirements Document (PRD) — Final
Project	Krozenda — Multi-Vendor B2B + B2C Marketplace
Client	[Legal Entity Name Pvt. Ltd.]
Vendor	Appzeto — appzeto.com
Quote reference	#0001710, issued 27 July 2026
Committed timeline	90 working days (Phase 1)
Version	1.0 — Final for client sign-off
Date	[DD Month YYYY]
Status	Awaiting client approval on the decisions in Part E §44

§ Document control

Purpose of this document

This PRD is the single functional reference for the Krozenda build. It states what will be built, how each part behaves, what the business rules are, and how completion will be judged. Where the scope has a boundary, this document draws it explicitly. Where a decision is still open, it is recorded in the register at §44 rather than assumed away.

Every functional requirement carries a stable identifier — `FR-M07-04` is the fourth requirement of Module 07 — so that requirements can be referenced unambiguously in test plans, change requests and acceptance sign-off.

Priority key

Priority	Meaning
P0	Contracted Phase 1 scope. Blocking for launch. Must be delivered and accepted within the 90 working days.
P1	Contracted Phase 1 scope, but degradable — the platform can launch with a documented manual workaround if a genuine blocker arises. Deferral requires written client agreement.
P2	Phase 2. Explicitly deferred, separately scoped and priced. Documented here so the architecture accommodates it without redevelopment.
ADD	Scope addition. Not in the agreed Phase 1 scope or the original SOW. Specified here because the client requested it. Requires a written change request covering cost and timeline before any work begins.

Source of authority

This PRD is derived from the agreed project context document and the quotation it implements. Where this PRD and the signed Scope of Work conflict, **the Scope of Work prevails** and the conflict must be raised as a change request. Nothing in this document silently expands what has been contracted.

Revision history

Version	Date	Author	Summary
1.0	[DD MMM YYYY]	Appzeto	Initial complete PRD covering all Phase 1 modules, flows and acceptance criteria.
[]	[]	[]	[Client review comments incorporated]

Sign-off

Role	Name	Signature	Date
Client — Project Sponsor	[]		
Client — Technical Contact	[]		
Appzeto — Delivery Lead	[]		

How to read this document. Part A establishes what Krozenda is and who uses it. Part B fixes the architecture and data model. Part C — the bulk — specifies all twenty-three functional modules in build order. Part D describes each user-facing surface screen by screen. Part E covers delivery, testing, risk and the decisions still awaiting the client. If you read only one section, read §44 — the open decisions determine the critical path.

§ Contents

PART A — PRODUCT & SCOPE

- 1 Product overview and vision
- 2 The three business models
- 3 User roles and personas
- 4 RBAC and the permission model
- 5 Scope boundary
- 6 Assumptions and dependencies

PART B — ARCHITECTURE

- 7 System architecture
- 8 Technology stack
- 9 Data model
- 10 Integration layer
- 11 API conventions
- 12 Non-functional requirements

PART C — FUNCTIONAL MODULES

- M01 Authentication and identity
- M02 User and role management
- M03 Seller onboarding and KYC
- M04 Category and brand management
- M05 Product management
- M06 Inventory management
- M07 Pricing and commission engine
- M08 Search and discovery
- M09 Cart and wishlist
- M10 Checkout and order splitting

- M11 Payment
- M12 Order management
- M13 Fulfilment and shipping
- M14 Returns, replacement and RTO
- M15 Settlement and payouts
- M16 Wallet and store credit
- M17 Coupons, offers and discounts
- M18 Reviews and ratings
- M19 Notification engine
- M20 Invoicing and tax
- M21 Reports and analytics
- M22 CMS and merchandising
- M23 Audit, security and backup

PART D — SURFACES

- 37 Buyer app and website
- 38 Seller panel and app
- 39 Admin panel

PART E — DELIVERY

- 40 Phase plan and timeline
- 41 Acceptance criteria
- 42 Testing strategy
- 43 Deployment and handover
- 44 Open decisions register
- 45 Risk register
- 46 Client-owned dependencies
- 47 Glossary

PART A

Product & Scope

What Krozenda is, the three business models it runs simultaneously, the thirteen roles that use it, and the precise boundary of what is being built.

1 Product overview and vision

2 The three business models

3 User roles and personas

4 RBAC and the permission model

5 Scope boundary — in, out, and deferred

6 Assumptions and dependencies

1. Product overview and vision

1.1 What Krozenda is

Krozenda is a single integrated commerce platform that runs three distinct business models at the same time, sharing one product catalog, one shopping cart, one checkout, and one settlement engine. A customer browsing the platform sees a single coherent storefront; behind it, any given product may be supplied by an external dropshipping partner, sold by an independent marketplace seller, or owned outright by the platform operator.

The platform serves both retail consumers and business buyers. The same product can carry different prices for a retail customer, a dealer, a distributor and a wholesaler, with minimum order quantities enforced per tier. This dual B2C/B2B capability on shared infrastructure is the platform's central design challenge and its primary commercial differentiator.

1.2 Problem statement

Operators wanting to run dropshipping, a marketplace, and their own inventory typically end up with three disconnected systems: a dropshipping tool, a marketplace platform, and an inventory system. Stock is duplicated, orders are reconciled by hand, commission is calculated in spreadsheets, and vendor settlement is a monthly ordeal. Reporting across the three is impossible without manual consolidation.

Krozenda addresses this with a unified model in which:

- one catalog holds products from all three sources, distinguished by an ownership attribute rather than by living in separate systems;
- one cart can contain items from several sources and splits automatically at checkout into independently fulfilled, invoiced and settled sub-orders;
- one commission engine resolves rates across products, vendors, categories and companies with a deterministic precedence order; and
- one settlement engine pays every vendor through a single compliant mechanism, with a complete audit trail.

1.3 Product goals

Goal	Definition of success
Unified operations	An operator manages dropshipping, marketplace and own-stock business from one admin panel, with no manual reconciliation between them.
Vendor self-service	A seller onboards, lists products, manages orders and reads settlements without contacting platform staff for routine work.
Financial correctness	Every rupee collected is traceable to a sub-order, a commission figure, a settlement record and a tax line. Historical settlements are reproducible.
Regulatory defensibility	Payment aggregation, GST, TDS, DLT and data protection obligations are met by design rather than retrofitted.
B2B and B2C parity	A business buyer gets tiered pricing, MOQ enforcement and a tax invoice through the same flows a retail customer uses.
Extensibility	Phase 2 items — open APIs, RFQ, credit limits, further supplier integrations — can be added without redeveloping Phase 1.

1.4 Out-of-scope goals

To keep the 90-day commitment honest, the platform explicitly does *not* aim, in Phase 1, to be an ERP, an accounting system, a CRM, a POS, a subscription-commerce engine, or an internationalised multi-currency storefront. These are enumerated in §5.2.

1.5 Success metrics

Metric	Launch target	Measured by
Seller onboarding completion rate	[70%] of started applications	Applications reaching active storefront ÷ applications begun
Order-to-dispatch time	< [48] hours median	Vendor Accepted → AWB Generated
Settlement accuracy	100% reconciled	Nightly reconciliation exception count = 0
Checkout completion rate	> [60%]	Payments captured ÷ checkouts initiated
API p95 latency	< 500 ms	Application performance monitoring
Platform uptime	99.5%	Monthly availability measurement

2. The three business models

Everything in this document is shaped by the fact that three models coexist. The `businessModel` attribute on a product, and on the sub-order derived from it, determines who fulfils, who invoices, who is paid, and who carries liability.

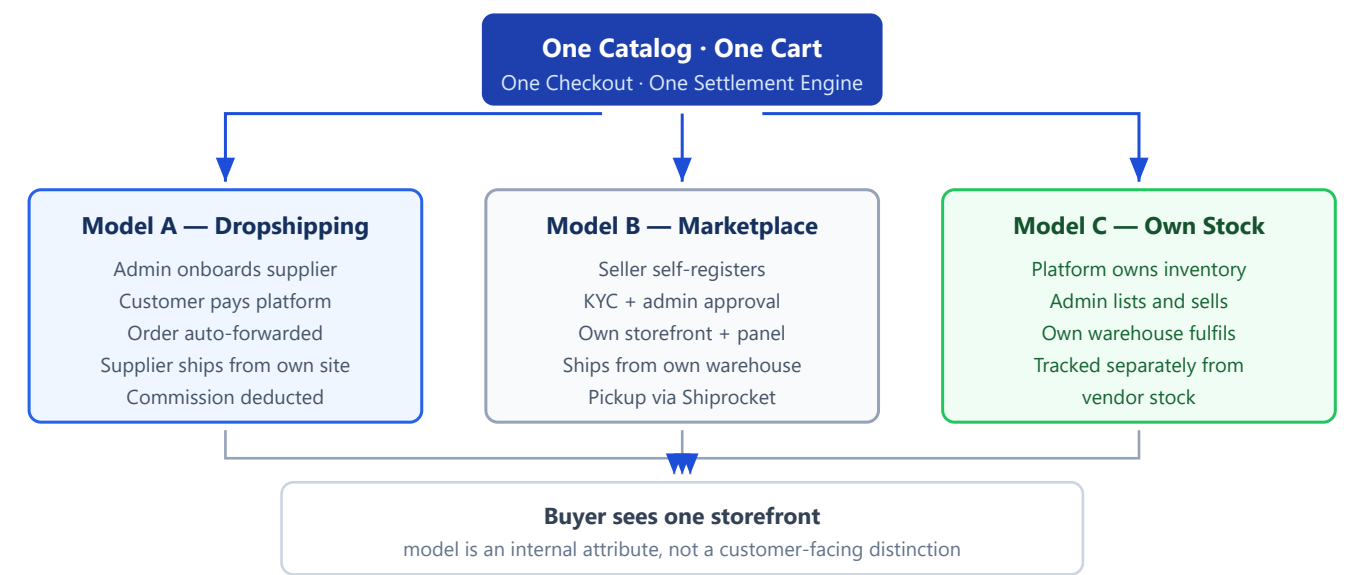


Figure 2.1 — Three business models sharing one commerce spine.

2.1 Model A — Direct Dropshipping

The Admin onboards external suppliers — companies, manufacturers, wholesalers, traders, dealers and retail shops. Their products appear in the platform catalog. The customer pays into the platform, the order is automatically forwarded to the supplier, and the supplier packs and ships from its own address, uploading tracking. The platform deducts commission and settles the remainder.

Attribute	Behaviour
Supplier acquisition	Admin-initiated only in Phase 1. Supplier self-registration is deferred to Phase 2.
Catalog source	Manual entry, CSV/Excel import, or API sync from a supplier adapter.
Supplier sync	Capped at two supplier integrations in Phase 1. Each further integration is bespoke, separately priced work.
Fulfilment	Supplier ships from its own premises; pickup scheduled through Shiprocket against the supplier's registered pickup address.
Settlement	Razorpay Route transfer to the supplier's linked account, net of commission.

2.2 Model B — Marketplace

Any seller may self-register, upload KYC documents, and wait for Admin approval. On approval the seller receives a storefront and access to the Seller Panel on both web and mobile, managing its own products, stock, pricing, orders and settlements. Fulfilment is from the seller's registered warehouse, with pickup arranged through Shiprocket.

2.3 Model C — Own Stock

The platform owner lists and sells platform-owned inventory, fulfilled from the platform's own warehouse. Inventory, orders, shipping and sales reports for own stock must be trackable **separately** from vendor stock — this is a reporting requirement that shapes the inventory data model, which carries an explicit owner bucket rather than inferring ownership from the product record.

2.4 Model comparison

Dimension	Dropshipping	Marketplace	Own Stock
Who onboards	Admin	Seller self-registers	n/a
KYC required	Yes	Yes	No
Owns inventory	Supplier	Seller	Platform
Ships from	Supplier premises	Seller warehouse	Platform warehouse
Route transfer	Yes	Yes	No — funds retained
Commission applies	Yes	Yes	No
Seller of record	[Open — §44 D2]	Seller	Platform
Inventory tracked	Supplier-reported	Seller-managed	Platform-managed

3. User roles and personas

Thirteen roles are supported. Each has a configurable permission set — permissions are data, never hardcoded — and each lands on one of three surfaces: the Admin Panel, the Seller Panel, or the Buyer app and website.

3.1 The thirteen roles

#	Role	Surface	Description and notes
1	Super Admin	Admin Panel	Full unrestricted access. Can create and remove Admins. Actions written to an immutable audit trail. At least one must always exist.

#	Role	Surface	Description and notes
2	Admin	Admin Panel	Scoped by an assigned permission set. Cannot escalate its own permissions.
3	Staff	Admin Panel	Narrow operational permissions — typically order processing or KYC review only.
4	Company	Seller Panel	Registered business entity selling on the platform.
5	Manufacturer	Seller Panel	Produces and sells its own goods.
6	Vendor / Seller	Seller Panel	Marketplace seller under Model B.
7	Dropshipping Partner	Seller Panel	Supplier under Model A. Admin-onboarded in Phase 1.
8	Trader	Seller Panel / Buyer	Dual capability — can sell and buy on one account.
9	Dealer	Seller Panel / Buyer	Dual capability. Dealer price tier when buying.
10	Distributor	Seller Panel / Buyer	Dual capability. Distributor price tier when buying.
11	Wholesaler	Seller Panel / Buyer	Dual capability. Wholesale price tier when buying.
12	Retail Customer	Buyer app / web	B2C. Standard retail pricing, no MOQ.
13	B2B Buyer	Buyer app / web	Tiered pricing access, MOQ enforcement, tax invoice.

Architectural requirement — do not model role as a string. Roles 8 to 11 can hold *both* a seller and a buyer capability on a single account. A user record therefore carries `roles[]` and `capabilities[]` arrays, never a single `role` field. Every authorisation check must ask "does this user hold this capability" rather than "is this user this role". Getting this wrong at the data layer forces a painful migration later, because dual-capability accounts cannot be represented at all.

3.2 Personas

Persona 1 — Platform Operations Manager (Admin)

Runs the marketplace day to day. Reviews seller KYC applications, approves products and categories, monitors orders that have stalled, handles escalated returns, configures commission, and reads the daily settlement report. Needs bulk actions, clear queues of pending work, and a reason field on every rejection so decisions are defensible later.

Persona 2 — Marketplace Seller

A small business owner listing a few hundred SKUs. Uploads products by CSV rather than one at a time, checks new orders on a phone during the day, prints packing slips in a batch each evening, and cares more about when money arrives than about anything else in the panel. Needs a settlement view that answers "when will I be paid and how much" without support.

Persona 3 — Dropshipping Partner

A wholesaler whose catalog is synced by adapter. Rarely logs in; receives forwarded orders, ships them, uploads tracking. Needs order forwarding to be reliable and settlement to be automatic. Failure mode to design against: stock drift between their system and ours causing accepted orders they cannot fulfil.

Persona 4 — Retail Customer

Buys occasionally on a phone. Cares about price, delivery date, and whether a return is possible. Will abandon checkout if it takes more than a few steps or if delivery to their PIN code turns out to be unavailable only at the last step.

Persona 5 — B2B Buyer (Dealer / Distributor)

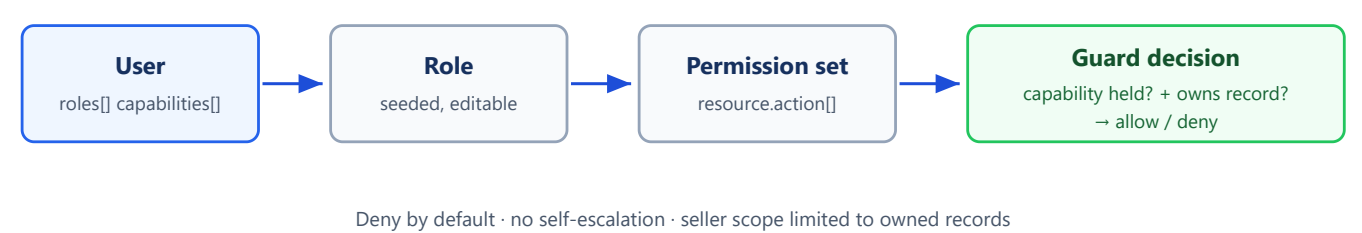
Buys in bulk at a negotiated tier, needs a GST invoice with the correct GSTIN, and repeats similar orders regularly. Values reorder, saved addresses, MOQ visibility and clear tier pricing far above browsing or discovery features.

4. RBAC and the permission model

4.1 Principles

- **Permissions are data.** Stored as records, seeded at deployment, editable by Super Admin without a code change.
- **Capability-based checks.** Authorisation asks whether the user holds a named capability, not which role they are.
- **Deny by default.** Absence of an explicit grant is a denial.
- **No self-escalation.** No role may grant itself a permission it does not already hold.
- **Ownership scoping.** Seller-surface permissions are additionally scoped to records the seller owns — a seller with `order.read` reads only its own sub-orders.

Permissions are records, not code. Changing what a role can do is configuration.



Deny by default · no self-escalation · seller scope limited to owned records

Figure 4.1 — Permission resolution chain.

4.2 Permission namespace

Permissions are named `resource.action`. The resource list follows the data model; actions are drawn from a fixed verb set so the namespace stays predictable.

Verb	Meaning
<code>read</code>	View the record or list.
<code>create</code>	Create a new record.
<code>update</code>	Modify an existing record.
<code>delete</code>	Soft-delete or deactivate.
<code>approve</code>	Move a record through an approval gate — KYC, product, category, brand, return.
<code>export</code>	Extract data as CSV or PDF. Separated from <code>read</code> because bulk extraction is a distinct risk.
<code>configure</code>	Change system settings — commission rules, templates, policies.

4.3 Indicative permission matrix

Abbreviated; the seeded matrix covers every resource. **Y** = full, **O** = own records only, **—** = denied.

Resource	Super Admin	Admin	Staff	Seller	Buyer
users	Y	Y	read	—	own profile
roles & permissions	Y	read	—	—	—
sellers / KYC	Y	Y	read, approve	O	—
categories & brands	Y	Y	read	read	read
products	Y	Y	read, approve	O	read
inventory	Y	Y	read	O	—
commission rules	Y	configure	—	read own	—
orders / sub-orders	Y	Y	read, update	O	own
settlements	Y	Y	read	read own	—
returns	Y	Y	approve	O	own
coupons & offers	Y	Y	read	—	—
CMS & banners	Y	Y	—	—	—
reports & analytics	Y	Y	read	own	—
audit logs	Y	read	—	—	—
backup & restore	Y	—	—	—	—

4.4 RBAC requirements

ID	Requirement	Pri
FR-RBAC-01	The system stores roles and permissions as database records seeded at deployment, with all thirteen roles present.	P0
FR-RBAC-02	A user record carries <code>roles[]</code> and <code>capabilities[]</code> arrays; no single-role string field exists anywhere in the schema.	P0
FR-RBAC-03	Every API endpoint is guarded by a declared capability requirement; an endpoint with no declaration is denied by default.	P0
FR-RBAC-04	Seller-surface endpoints additionally scope results to records owned by the authenticated seller.	P0
FR-RBAC-05	Super Admin can create, edit and assign permission sets from the Admin Panel without a deployment.	P0
FR-RBAC-06	No role can grant itself or a peer a capability it does not hold.	P0
FR-RBAC-07	A user holding both seller and buyer capabilities can switch surface without logging out and re-authenticating.	P1
FR-RBAC-08	Every permission change is written to the audit log with actor, before state, after state, timestamp and IP.	P0

ID	Requirement	Pri
FR-RBAC-09	The system prevents removal or demotion of the last remaining Super Admin.	P0

5. Scope boundary

5.1 In scope — Phase 1

The contracted Phase 1 delivery comprises the twenty-three modules in Part C, the three surfaces in Part D, and four app binaries — Buyer Android, Buyer iOS, Seller Android, Seller iOS — all produced from React Native, plus the responsive React web platform.

5.2 Explicitly excluded — not being built

The following appear in the original Scope of Work but are **not** part of this delivery. They will not be built, scaffolded, stubbed or referenced in the codebase.

Excluded item	Note
Tally integration — ledgers, voucher export, GST ledger mapping, inventory sync	Excluded entirely
Zoho Books integration	Excluded entirely
WhatsApp notifications / WhatsApp Business API	Excluded entirely
AI chatbot	Excluded entirely
Automated KYC verification APIs (PAN / Aadhaar / GST / FSSAI)	Human review only — see M03
Bank account penny-drop verification	See risk R-04
Cash on Delivery	Online payment only
Multi-language / multi-currency	INR and English only
ERP / CRM / POS / Franchise / Membership / Subscription modules	Excluded entirely
Firebase Phone Auth, Firebase Auth, Firestore, Firebase Storage, Remote Config	FCM push only
Native Android (Kotlin/Java) or native iOS (Swift/ObjC) code	React Native only
Post-delivery support, warranty period, AMC	Separate engagement
Data migration from any existing system	Not scoped

5.3 Deferred to Phase 2

Deferred item	Reason
Open APIs — Product, Company, Inventory, ERP, Shipping, Accounting, Order	Seven documented public APIs is a substantial standalone deliverable with no consumer on day one
B2B RFQ workflow	Distinct multi-step workflow module

Deferred item	Reason
B2B Purchase Orders	Depends on RFQ
B2B Credit Limit	Requires a credit ledger and exposure tracking
Supplier integrations beyond two	Each is bespoke adapter work, priced per supplier
Supplier self-registration as Dropshipping Partner	Admin adds suppliers manually in Phase 1
External partners selling platform products via API	Future

Architectural obligation. The codebase must be modular enough that every Phase 2 item can be added without redevelopment. Integration layers stay behind interfaces — payment provider, logistics provider, notification channel, supplier adapter — so a new implementation is a new class, not a new codebase.

5.4 Scope additions requested outside the SOW

Item	Status	Note
Wallet / store credit (M16)	ADD	Specified in full at M16 at the client's request. Not in the SOW, not in the 90-day commitment. Carries RBI prepaid-instrument implications — see M16.2. Requires a written change request before any work starts.
Automated salary payouts	ADD	Disbursement machinery is specified in M15.8. Statutory payroll computation — PF, ESI, PT, s.192 TDS, Form 16 — is <i>not</i> included and would be a separate engagement.

6. Assumptions and dependencies

6.1 Assumptions

ID	Assumption	If false
A-01	The client's Razorpay account has Route provisioned before Phase 0 ends.	All settlement work (M15) is blocked; critical path slips day for day.
A-02	TRAI DLT registration, sender ID and SMS templates are approved before UAT.	OTP cannot be delivered; login is untestable end to end.
A-03	The client supplies final legal policy text — Privacy, T&C, Shipping, Return, Vendor Agreement.	Policy acceptance flow ships with placeholder text and cannot go live.
A-04	Brand assets, logo and product content are supplied by the client.	UI ships against placeholder brand tokens.
A-05	Non-functional targets in §12 are accepted as stated.	Performance testing has no pass criteria.
A-06	Catalog is English and pricing is INR only.	Multi-language/currency is an excluded item and a change request.
A-07	At most two supplier API integrations are required in Phase 1.	Each additional adapter is separately priced.
A-08	The client provides one named decision-maker able to resolve §44 items within 48 hours.	Blocked decisions accumulate and compress the hardening phase.

6.2 Client-owned dependencies

Development is blocked without these. Each needs a named owner and a committed date — see §46 for the tracking table.

Dependency	Owner
Razorpay account with Route enabled	Client
Razorpay MDR and transaction charges agreed	Client
Shiprocket account + per-vendor pickup locations	Client
TRAI DLT registration, Sender ID, SMS template approval	Client
SMTP credentials + SPF / DKIM / DMARC DNS records	Client
Firebase project ownership	Client
Domain and hosting	Client
Apple Developer account (\$99/yr) + Google Play Console	Client

Dependency	Owner
Legal policy text — Privacy, T&C, Shipping, Return, Vendor Agreement	Client
Seller-of-record and GST TCS/TDS position	Client
Logo, brand assets, product content	Client
UI/UX design and design system	Appzeto

6.3 Compliance obligations

Regime	Obligation on the platform
RBI Payment Aggregator framework	Vendor funds cannot be held and disbursed by the platform without a PA licence. Razorpay Route is the compliance path. See M11 and M15.
GST	TCS under s.52 and TDS under s.194-O apply to marketplace operators. Invoice structure depends on the seller-of-record decision.
TRAI DLT	Every transactional SMS template must be pre-registered. Templates must be data, not code.
Consumer Protection (E-Commerce) Rules 2020	Grievance officer, published return/refund terms, and the No-Return reconciliation flagged in M14.
IT Act / DPDP Act 2023	Encryption of KYC documents and PII, breach handling, consent records, data principal rights.
FSSAI	Required for food-category sellers; enforced at category approval.
Legal Metrology	Mandatory declarations on packaged commodity listings.

PART B

Architecture

The system shape, the technology stack, the data model every module depends on, the integration boundaries, and the non-functional targets the build must hit.

7 System architecture

8 Technology stack

9 Data model and core collections

10 Integration layer — the five integrations

11 API conventions

12 Non-functional requirements

7. System architecture

CLIENT SURFACES



REST API — Express · JWT auth · RBAC guard · rate limit · Zod validation
versioned · OpenAPI contract · every endpoint declares a required capability

DOMAIN MODULES



INTEGRATION INTERFACES — swap the implementation, never the caller



Inbound webhooks: payment.captured · transfer.processed · tracking events · payout status

Figure 7.1 — System architecture. Domain modules never import a vendor SDK directly; they depend on the interface layer.

7.1 Architectural principles

Principle	Consequence for the build
Modular by domain	Code is organised by business domain — auth, seller, catalog, order — not by technical layer. Each module owns its controllers, services, validators and constants.
Integrations behind interfaces	No module imports the Razorpay, Shiprocket or FCM SDK directly. They depend on <code>PaymentProvider</code> , <code>LogisticsProvider</code> , <code>NotificationChannel</code> and <code>SupplierAdapter</code> . This is what makes Phase 2 additions non-destructive and makes the whole system testable without live third-party calls.
MongoDB is the system of record	No business state lives in a third-party service. Razorpay and Shiprocket hold their own records; ours are authoritative for our domain.
Financial values are snapshotted	Price, commission and GST rate are written onto the order at the moment of placement and never recomputed from live configuration.
Server-authoritative amounts	Every payable figure is computed server-side from the database. A price submitted by a client is never trusted.
Fail closed on money	Where a settlement or payout condition cannot be confirmed, the system withholds rather than releases, and raises an exception for human review.
Shared constants across the wire	Role keys, status keys and business-model keys are defined once and mirrored in the frontend config, because they cross the wire as plain strings.

7.2 Code organisation

```
backend/src
  config/          constants (roles, statuses, business models), env, db
  models/          Mongoose schemas – the core collections in §9
  modules/         one folder per domain, each with:
    <domain>/      controllers/ services/ validators/ routes.js constants.js
    admin/ auth/   seller/ dropshipping-partner/ user/ vendor-shared/
  integrations/    the four interface families
    payment/       PaymentProvider.js RazorpayProvider.js
    logistics/     LogisticsProvider.js ShiprocketProvider.js
    notification/  NotificationChannel.js FcmChannel.js SmsIndiaHubChannel.js SmtplibChannel.js
    supplier-adapters/ SupplierAdapter.js index.js
  middlewares/     auth guard · RBAC guard · upload · error handler · rate limit
  routes/          route registration
  scripts/         seeders (roles, permissions, reference data)
  lib/             shared helpers

frontend/src
  modules/         admin/ auth/ seller/ dropshipping-partner/ user/
    each with:     pages/ components/ controllers/ services/ schemas/
  components/      common/ layout/ ui/ ← shared design system
  config/          constants mirrored from backend
  lib/             api client, query client, helpers
```

vendor-shared exists because a Marketplace Seller and a Dropshipping Partner use largely the same surface. Shared behaviour lives there; only genuinely divergent behaviour is duplicated into the two role-specific modules.

7.3 Request lifecycle

```
request
→ rate limiter      reject if over threshold for this IP / user / endpoint class
→ JWT verification  decode, check expiry, load user with roles[] + capabilities[]
→ RBAC guard        endpoint declares required capability; deny if absent
→ ownership scope   seller surface: constrain query to records owned by this seller
→ Zod validation    parse and coerce body / params / query; reject on failure
→ controller        thin – no business logic
→ service           business logic, transactions, snapshotting
→ model             Mongoose
→ audit log         on privileged mutations: actor, entity, before, after, ip, ts
→ response envelope consistent shape (§11)
```

8. Technology stack

MERN only. This is a JavaScript end-to-end project. No Kotlin or Java for Android, no Swift or Objective-C for iOS, and no second backend language or non-JS runtime anywhere in the stack. The four app binaries are produced from React Native, sharing logic and components with the React web frontend wherever possible.

Layer	Technology	Notes
Backend runtime	Node.js, Express 5	ES modules throughout

Layer	Technology	Notes
Database	MongoDB with Mongoose	System of record
Validation	Zod	Shared schema shapes between backend and frontend
Auth	JWT + refresh tokens, bcrypt	Own implementation — not Firebase Auth
Security middleware	helmet, cors, express-rate-limit, cookie-parser	Applied globally
Uploads	multer → object storage	KYC encrypted at rest
Web frontend	React 18, Vite, React Router 6	Responsive; no separate mobile web build
Styling	Tailwind CSS	Brand tokens only — JSX never references raw hex
State / data	Zustand, TanStack Query	Server state via Query; UI state via Zustand
Forms	React Hook Form + Zod resolver	
Mobile apps	React Native	Buyer Android/iOS, Seller Android/iOS
Push	Firebase Cloud Messaging	FCM only — no other Firebase product
Payments	Razorpay + Route	RazorpayX Payouts for M15.8
Logistics	Shiprocket	
SMS + OTP	SMS India Hub	All OTP — never Firebase Phone Auth
Email	SMTP, queued sender	SPF/DKIM/DMARC client-owned
Search	[Atlas Search — pending D3]	See \$44 decision D3

9. Data model

9.1 Core collections

These must be established before feature work begins. The list is foundational, not exhaustive.

users	roles[], capabilities[], profile, auth, status
sellers	user ref, KYC docs, approval state, store config, razorpay_linked_account_id, business model, pickup addresses
kyc_documents	type, file ref, status, reviewer, review notes, version
categories	tree, approval state
brands	approval state
products	owner (seller admin), type, variants[], sku, barcode, hsn, gst_rate, moq, approval state, source (manual csv api), supplier_ref
price_tiers	product ref, buyer_role, price, min_qty
inventory	product/variant, owner bucket (vendor own_stock), qty, reserved qty
commission_rules	scope (product vendor category company default), type, value
carts	line items with resolved price + tier snapshot
orders	parent order, buyer, payment ref, totals
sub_orders	parent ref, vendor ref, model, status, commission snapshot, invoice ref
shipments	sub_order ref, awb, courier, tracking events[]
invoices	sub_order ref, gst breakup, seller of record
settlements	sub_order ref, gross, commission, net, route transfer id
returns	sub_order ref, reason, evidence[], resolution
notifications	channel, template ref, recipient, status
audit_logs	actor, entity, before/after, timestamp, ip
policy_acceptances	user, policy type, version, timestamp, ip

9.2 Entity relationships

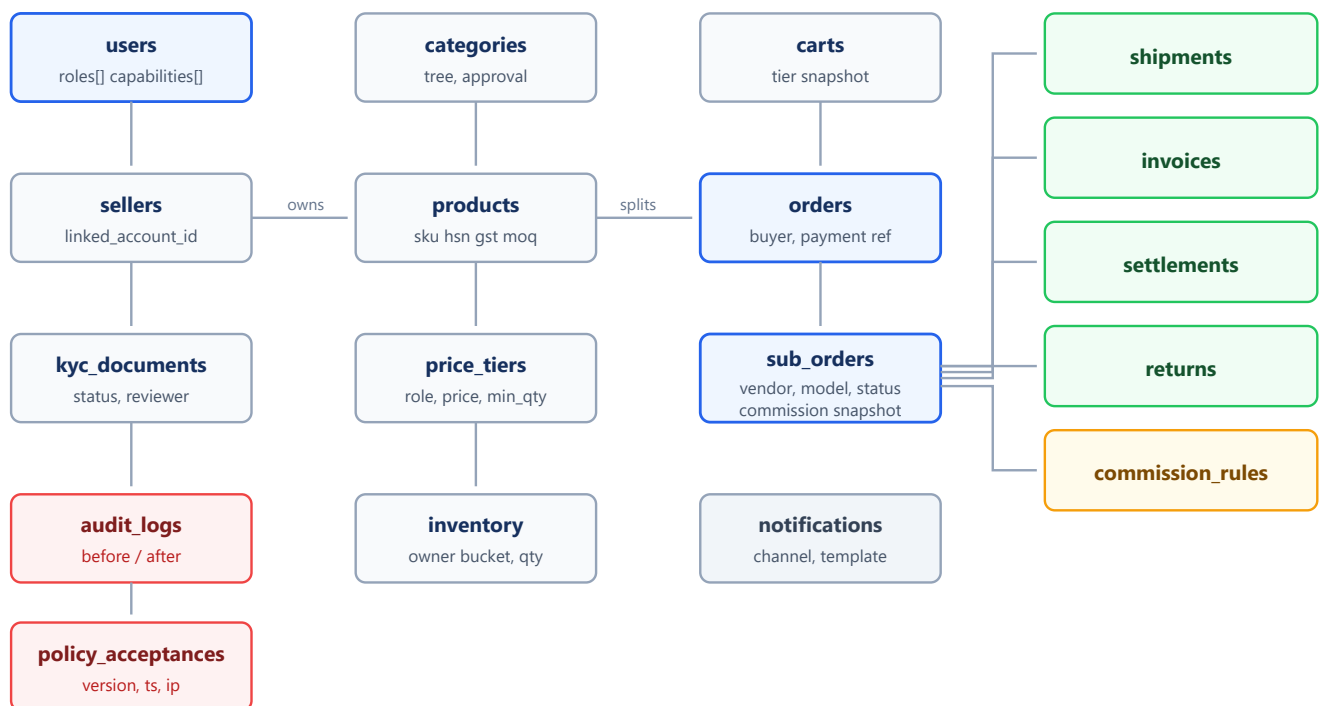


Figure 9.1 — Core entity relationships. Blue = transactional spine, green = per-sub-order artefacts, red = compliance records.

9.3 Data model rules

ID	Requirement	Pri
FR-DM-01	A user record carries <code>roles[]</code> and <code>capabilities[]</code> ; no single-role string field exists.	P0
FR-DM-02	Sub-orders carry a commission snapshot written at order placement and never recomputed.	P0

ID	Requirement	Pri
FR-DM-03	Cart lines carry the resolved price and the tier that produced it, so a later tier change does not silently reprice an open cart.	P0
FR-DM-04	Inventory carries an explicit owner bucket — <code>vendor</code> or <code>own_stock</code> — so own-stock reporting never requires inference.	P0
FR-DM-05	Invoices record the seller of record explicitly rather than deriving it at render time.	P0
FR-DM-06	Audit log entries are append-only; no application path updates or deletes them.	P0
FR-DM-07	Policy acceptances record the policy type, version, timestamp and IP, and are retained per the Privacy Policy schedule.	P0
FR-DM-08	Every financial amount is stored as an integer in paise, never as a floating-point rupee value.	P0
FR-DM-09	Soft deletion is used for catalog and user records; hard deletion is reserved for data-subject erasure requests.	P1

10. Integration layer

Exactly five integrations are in scope. No other third-party service will be introduced.

10.1 Razorpay — payments and settlement

Aspect	Specification
Methods	UPI, credit card, debit card, net banking, wallet
Payment mode	Online only. No COD.
Settlement mechanism	Razorpay Route — mandatory. Every vendor is a Route linked account with its own KYC.
Flow	Customer → Razorpay → Platform (commission) + Linked Account (vendor share)
Implementation	Route transfers and holds. Not a home-grown payout ledger.
Must handle	Payment capture, webhook signature verification, refunds, partial refunds, transfer reversal

Compliance constraint. Collecting vendor funds into the platform account and disbursing them manually is not permissible under the RBI Payment Aggregator framework without a PA licence. Route is the compliance path, not a convenience. See M11 and M15.

10.2 Shiprocket — logistics

- AWB generation, courier allocation, shipping label, pickup scheduling, real-time tracking, delivery status webhooks.
- Vendor pickup *and* platform warehouse pickup addresses.
- Client owns the Shiprocket account, sub-accounts and per-vendor pickup location setup.
- **RTO handling must be built** — see M14.

10.3 Firebase — push notifications only

FCM and nothing else. Do not use Firebase Phone Auth — all OTP goes through SMS India Hub. Do not use Firebase Auth, Firestore, Firebase Storage or Remote Config. MongoDB is the system of record and authentication is our own JWT/refresh implementation.

10.4 SMS India Hub — SMS and all OTP

- Transactional SMS and **all** OTP delivery — login, registration, order alerts.
- Because Firebase Phone Auth is excluded, OTP generation, hashing, expiry, retry limits and rate limiting are **our responsibility**, built into the auth service.
- **DLT dependency (client-owned)**: TRAI DLT entity registration, sender ID approval, and pre-registration of *every* template. Takes 3–10 working days and templates cannot be changed freely after approval.
- Templates are built as configurable records so DLT-approved text can be swapped without a code deploy.

10.5 SMTP — email

- Order confirmations, invoices, email verification links, seller notifications.
- **Client-owned infrastructure dependency**: SPF, DKIM and DMARC records on the sending domain. Without these, deliverability fails at volume.
- Implemented as a queued sender with retry — never sent synchronously inside the request cycle.

10.6 Supplier adapters

Supplier sync — stock, price, image, description — is **capped at two supplier integrations in Phase 1**. Each further integration is bespoke work, priced separately. A pluggable adapter interface ensures a new supplier is a new adapter class, not a new codebase.

11. API conventions

11.1 Response envelope

```
success  { "success": true,  "data": <payload>, "meta": { pagination, ... } }
failure  { "success": false, "error": { "code": "STRING_CODE",
                                         "message": "human readable",
                                         "details": [ ... field errors ... ] } }
```

11.2 Conventions

Concern	Convention
Versioning	Path-versioned, <code>/api/v1/...</code> . Breaking changes require a new version.
Auth	Bearer JWT access token; refresh token in an httpOnly cookie.
Pagination	Cursor or page/limit, returned in <code>meta</code> . Default limit 20, maximum 100.
Filtering & sorting	Query parameters with an explicit allowlist per endpoint — never arbitrary field injection into the query.
Idempotency	Required on order placement, payment confirmation and payout creation.
Errors	Machine-readable code plus human message. Validation errors return per-field detail.
Rate limits	Tighter classes on auth, OTP and order placement than on catalog reads.
Contract	OpenAPI specification maintained in Phase 0 so frontend teams work against it before endpoints exist.
Timestamps	ISO-8601 UTC on the wire; rendered in IST on surfaces.
Money	Integer paise on the wire; formatted at the presentation layer only.

12. Non-functional requirements

These targets were absent from the Scope of Work. The figures below are working targets proposed by Appzeto and **must be confirmed by the client.** They become the pass criteria for performance testing in §42; without confirmation, performance testing has nothing to test against.

12.1 Performance and capacity

Metric	Working target	Verification
Concurrent users	1,000	Load test at 1.5× target
Catalog size	100,000 SKUs	Seeded dataset, search and listing latency measured at volume
Orders per day	5,000	Sustained-throughput test including settlement job
API p95 response	< 500 ms	APM under representative load
Uptime	99.5%	Monthly availability

12.2 Security requirements

ID	Requirement	Pri
FR-NFR-01	SSL/TLS enforced on every surface and every API call; HTTP redirects to HTTPS.	P0
FR-NFR-02	OTP authentication for all users; two-factor authentication additionally available for Admin and Seller accounts.	P0
FR-NFR-03	KYC documents and PII encrypted at rest.	P0
FR-NFR-04	Rate limiting, anti-spam and anti-fraud checks on order placement and on all auth endpoints.	P0
FR-NFR-05	Daily automated backup plus off-site cloud backup, with a tested restore procedure.	P0
FR-NFR-06	Login alerts sent to the account owner on a new device or unusual location.	P1
FR-NFR-07	Audit logs written on all privileged actions, append-only.	P0
FR-NFR-08	Passwords stored only as salted one-way hashes; never recoverable.	P0
FR-NFR-09	Secrets held in environment configuration, never in source control; test and live keys never coexist in one environment.	P0
FR-NFR-10	Security headers, input validation and protection against injection, XSS and CSRF applied globally.	P0

12.3 Usability and accessibility

ID	Requirement	Pri
FR-NFR-11	All web surfaces are responsive from 320 px to desktop widths.	P0
FR-NFR-12	Interactive elements meet a minimum 44×44 px touch target on mobile.	P1
FR-NFR-13	Text and interactive elements meet WCAG AA contrast ratios.	P1
FR-NFR-14	Every destructive action requires confirmation and states what will be affected.	P0
FR-NFR-15	Every error message states what went wrong and what the user can do next.	P1

PART C

Functional Modules

Twenty-three modules in build order. Each states its purpose, actors, functional requirements, business rules, flows, edge cases and acceptance criteria.

M01–M06 Identity, sellers, catalog, products, inventory

M07–M12 Pricing, search, cart, checkout, payment, orders

M13–M18 Fulfilment, returns, settlement, wallet, offers, reviews

M19–M23 Notifications, invoicing, reports, CMS, security

Authentication and Identity

Registration · OTP · JWT sessions · 2FA · password management · policy acceptance

M01.1 Purpose

Establishes who a user is and keeps that assertion valid across web and mobile. Because Firebase Phone Auth is excluded, the entire OTP lifecycle — generation, hashing, expiry, retry limiting and rate limiting — is built in-house and delivered through SMS India Hub. This module is foundational: every other module depends on it, and its exit criteria gate Phase 0.

M01.2 Actors

All thirteen roles. Retail customers and B2B buyers register themselves; sellers register and then enter the KYC flow in M03; Admin and Staff accounts are created by a Super Admin, never self-registered.

M01.3 Registration and login flow

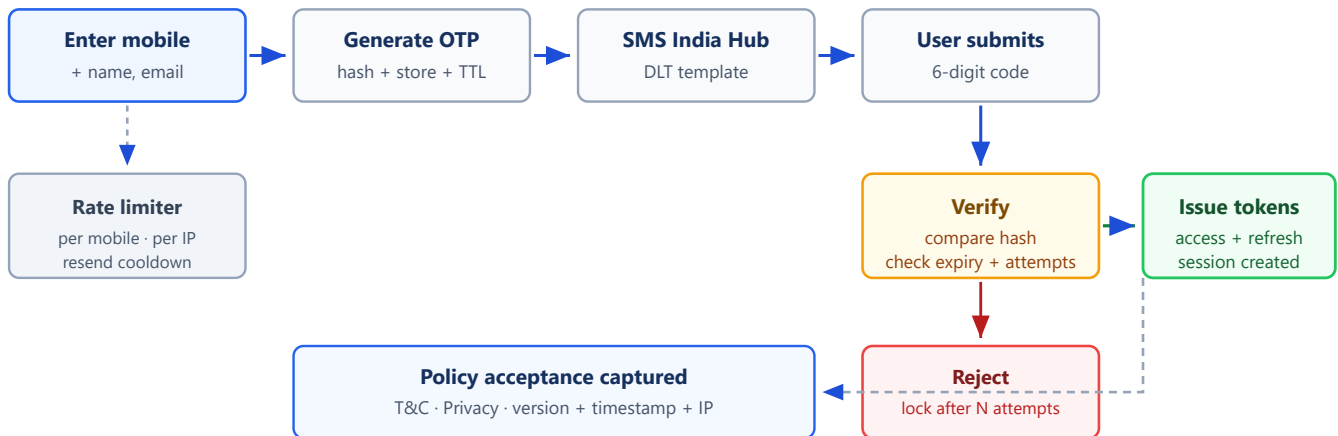


Figure M01.1 — OTP registration and login. OTP is hashed at rest; the plaintext code exists only in the SMS.

M01.4 Functional requirements

ID	Requirement	Pri
FR-M01-01	A buyer registers with name, mobile number and email. Mobile is verified by OTP; email by a verification link.	P0
FR-M01-02	OTP is a 6-digit numeric code with a configurable TTL, default [5] minutes.	P0
FR-M01-03	OTP is stored only as a hash. The plaintext code exists solely in the outbound SMS and is never logged.	P0
FR-M01-04	A maximum of [5] verification attempts per OTP; on exhaustion the OTP is invalidated and a new request is required.	P0
FR-M01-05	Resend is rate-limited with a cooldown of [60] seconds and a cap of [5] requests per mobile number per hour.	P0

ID	Requirement	Pri
FR-M01-06	Login is supported by mobile + OTP, and by email + password for users who have set one.	P0
FR-M01-07	Successful authentication issues a short-lived JWT access token and a long-lived refresh token; refresh tokens are rotated on use.	P0
FR-M01-08	A user can view active sessions and revoke any of them, including all other devices at once.	P1
FR-M01-09	Two-factor authentication is available for Admin, Staff and Seller accounts and can be enforced by policy per role.	P0
FR-M01-10	Password reset is by verified email link or mobile OTP; reset invalidates all existing sessions.	P0
FR-M01-11	Passwords are stored as bcrypt hashes with a per-user salt and are excluded from all query projections by default.	P0
FR-M01-12	At registration, acceptance of T&C and Privacy Policy is captured with policy version, timestamp and IP.	P0
FR-M01-13	Sellers additionally accept the Vendor Agreement, Shipping Policy and Return Policy, each recorded separately.	P0
FR-M01-14	Account status is one of <code>pending</code> , <code>active</code> or <code>suspended</code> ; only active accounts may transact.	P0
FR-M01-15	A login from a new device or unusual location triggers an alert to the registered email and mobile.	P1
FR-M01-16	Admin and Staff accounts are created only by a Super Admin; self-registration into those roles is impossible.	P0
FR-M01-17	A user holding both buyer and seller capabilities switches surface without re-authenticating.	P1
FR-M01-18	Mobile numbers and email addresses are unique across the platform where present.	P0

M01.5 Business rules

- **OTP delivery is single-channel.** All OTP goes through SMS India Hub. There is no fallback to email OTP and no Firebase path.
- **Templates are data.** SMS content is a configurable record referencing a DLT-approved template ID, so approved text can change without a deploy.
- **Enumeration resistance.** Login and reset responses do not reveal whether an identifier exists.
- **Session invalidation.** Password change, 2FA change and role change all invalidate existing sessions.

M01.6 Edge cases

Case	Handling
SMS gateway down	OTP request fails with a clear message and is not counted against the user's retry budget. Failures alert operations.
DLT template not yet approved	Delivery fails at the provider. Staging must run against a sandbox harness so login remains testable before approval lands.
User changes mobile number	Requires OTP verification on both the old and the new number before the change takes effect.
Same person, buyer and seller	One account with both capabilities — not two accounts. Enforced by the unique mobile constraint.
Refresh token replay	Rotation detects reuse of a consumed token and revokes the whole session family.

M01.7 Acceptance criteria

- A new user registers, receives an OTP, verifies, and reaches an authenticated state on both web and mobile.
- A permission-gated endpoint returns 403 before role assignment and 200 after — the Phase 0 exit criterion.
- Exhausting OTP attempts locks the code; a fresh request succeeds.
- Policy acceptance rows exist with correct version, timestamp and IP.
- 2FA can be enabled, enforced by policy, and used to log in.

MODULE 02 · PHASE 0

User and Role Management

User directory · role assignment · permission sets · suspension · impersonation

M02.1 Purpose

Gives Admin the tools to see, search and manage every account on the platform, and to configure what each role can do — without a developer or a deployment.

M02.2 Functional requirements

ID	Requirement	Pri
FR-M02-01	Admin can list, search and filter all users by role, capability, status, registration date and activity.	P0
FR-M02-02	Admin can view a user's full profile: contact details, roles, capabilities, addresses, order history, and — for sellers — KYC and settlement summary.	P0
FR-M02-03	Admin can assign and remove roles and capabilities on an account, subject to FR-RBAC-06.	P0
FR-M02-04	Admin can suspend and reinstate an account with a mandatory reason, notifying the user.	P0
FR-M02-05	Suspension immediately invalidates sessions and blocks new orders, while preserving historical records.	P0
FR-M02-06	Super Admin can create, edit, clone and delete permission sets, and assign them to roles.	P0
FR-M02-07	Admin can create Staff accounts with a chosen permission set.	P0
FR-M02-08	All user-management actions are written to the audit log with before and after state.	P0
FR-M02-09	Admin can export a filtered user list to CSV, gated behind a separate <code>export</code> permission.	P1
FR-M02-10	Admin can trigger a password reset or force 2FA enrolment on an account.	P1
FR-M02-11	Support impersonation ("view as user") is available to Super Admin only, is time-boxed, is banner-flagged in the UI, and is audit-logged. It never permits placing an order or changing payment details.	P2
FR-M02-12	A data-subject erasure request can be executed, removing personal data while retaining statutorily required financial records.	P1

M02.3 Business rules

- The last Super Admin cannot be deleted, demoted or suspended.
- Role changes take effect on the next token refresh at the latest; privileged changes force immediate re-authentication.
- A seller account cannot be deleted while it has unsettled sub-orders or open returns.
- Suspension is reversible; erasure is not, and requires a second confirmation naming the account.

Seller Onboarding and KYC

Registration · document upload · manual review · Route linked account · storefront activation

M03.1 Purpose

Takes a prospective seller from first registration to an active storefront capable of receiving orders and money. This module owns the human review workflow and the gating that prevents an unverified vendor from transacting.

Scope boundary — KYC is human, not automated. Automated verification APIs for PAN, Aadhaar, GST and FSSAI are explicitly excluded. The platform collects documents and an Admin reviews them by eye. Mobile OTP and email verification are the only automated checks. Nothing in this module should be described to end users as "verified by government records".

M03.2 Onboarding flow

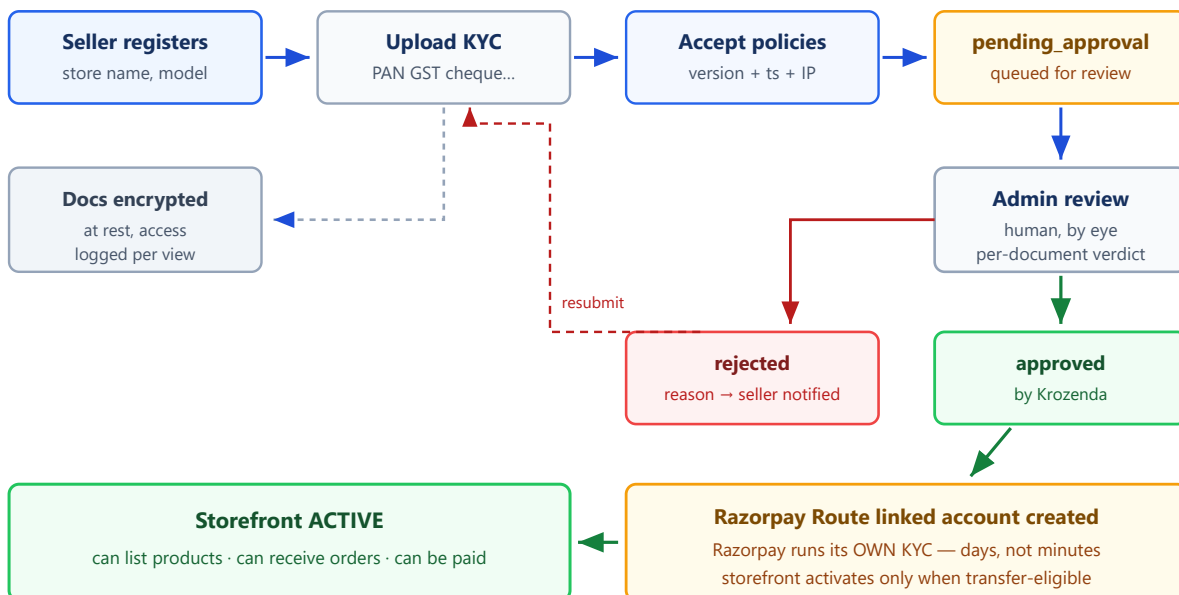


Figure M03.1 — Seller onboarding. Two independent approvals must both pass: Krozenda's, then Razorpay's.

M03.3 Documents collected

Document	When required	Notes
PAN card	Always	Of the business or the proprietor
Aadhaar or government photo ID	Always	Of the authorised signatory. Handled under its governing law; not used as a general identifier.
GST registration certificate	Always	Must match the legal name on PAN
FSSAI licence	Food categories	Enforced at category approval, not just at onboarding
Cancelled cheque / bank proof	Always	Must be in the business's name — see risk R-04
Address proof	Always	Registered and pickup addresses
Incorporation / partnership documents	Entities	Not required for sole proprietors

M03.4 Functional requirements

ID	Requirement	Pri
FR-M03-01	A seller self-registers with store name, business model, business type and authorised-person contact details.	P0
FR-M03-02	Dropshipping Partners are created by Admin. Supplier self-registration is Phase 2 and must not be built.	P0
FR-M03-03	Sellers upload KYC documents with a document type label; accepted formats and a size cap are enforced.	P0
FR-M03-04	Documents are stored encrypted at rest and served only through access-controlled URLs, never a public path.	P0
FR-M03-05	Admin reviews each document individually and records a per-document verdict with notes.	P0
FR-M03-06	Admin approves or rejects the application as a whole; rejection requires a reason, which is sent to the seller.	P0
FR-M03-07	A rejected seller can correct and resubmit; document versions are retained rather than overwritten.	P0
FR-M03-08	Seller status is one of <code>pending_approval</code> , <code>approved</code> , <code>rejected</code> , <code>suspended</code> .	P0
FR-M03-09	On approval, the system creates a Razorpay Route linked account and stores its identifier against the seller.	P0
FR-M03-10	The Seller Panel shows the linked-account KYC state distinctly from Krozenda's own approval state.	P0
FR-M03-11	A seller whose linked account is not transfer-eligible cannot be assigned orders. The system blocks assignment rather than accruing unpayable settlements.	P0
FR-M03-12	Digital acceptance of Vendor Agreement, Government Rules, Privacy Policy, T&C, Shipping Policy and Return Policy is captured with timestamp, IP and policy version.	P0
FR-M03-13	Sellers register pickup and warehouse addresses, which flow to Shiprocket as pickup locations.	P0
FR-M03-14	Admin sees a work queue of pending applications sorted by age, with document counts and time-in-queue.	P0

ID	Requirement	Pri
FR-M03-15	Every KYC document view by a staff member is logged with actor, document and timestamp.	P0
FR-M03-16	Sellers are notified at each state change by email, SMS and push.	P0
FR-M03-17	Admin can request additional documents without rejecting the whole application.	P1
FR-M03-18	Re-verification can be triggered when a seller changes bank details, GSTIN or legal name.	P1

M03.5 Edge cases

Case	Handling
Krozenda approves but Razorpay rejects	Seller stays approved but non-transactable. Panel states the payment partner rejected verification and what to correct. Order assignment stays blocked.
Razorpay KYC takes a week	Expected. Status is surfaced honestly rather than shown as a generic "pending", to keep support load down.
Bank details mistyped	No penny-drop check exists (see R-04). Mitigation is a manual comparison of the cancelled cheque against the entered account number during review, recorded as a checklist item.
Seller changes bank account after approval	Requires re-verification and a new or updated linked account; settlements pause until it is transfer-eligible.
Duplicate application, same PAN	Flagged to the reviewer as a possible duplicate; not auto-rejected, because legitimate multi-store cases exist.

M03.6 Acceptance criteria

- A seller registers, uploads all documents, accepts all policies, and reaches `pending_approval`.
- Admin reviews and rejects with a reason; the seller sees the reason and resubmits; both document versions are retained.
- On approval, a linked account is created and its state is visible in the Seller Panel.
- Order assignment to a non-transfer-eligible seller is refused by the system.
- Policy acceptance rows exist for all six documents with versions and IPs.

MODULE 04 · PHASE 1A · DAYS 11–30

Category and Brand Management

Category tree · brand registry · approval chains · category-level rules

M04.1 Purpose

Provides the taxonomy every product hangs from, and the first two gates of the three-stage product approval chain. Categories also carry rules that constrain what may be listed under them — the FSSAI requirement for food is enforced here rather than being left to reviewer memory.

M04.2 Functional requirements

ID	Requirement	Pri
FR-M04-01	Categories form a tree of arbitrary depth with a practical limit of <code>[4]</code> levels.	P0
FR-M04-02	Admin can create, rename, reorder, move, activate and deactivate categories.	P0

ID	Requirement	Pri
FR-M04-03	A category carries an approval state; products can only be listed under an approved, active category.	P0
FR-M04-04	A category can require named licences — for example FSSAI — which the seller must hold before listing under it.	P0
FR-M04-05	A category can carry a default GST rate and HSN hint, used as a prefill when a product is created.	P1
FR-M04-06	A category can carry a default commission rule, participating in the resolution order in M07.	P0
FR-M04-07	Brands are a managed registry with an approval state; sellers select from approved brands or request a new one.	P0
FR-M04-08	A brand request captures the requesting seller and any authorisation document, and is approved or rejected with a reason.	P0
FR-M04-09	Deactivating a category with live products is blocked; Admin must first reassign or delist those products.	P0
FR-M04-10	Category and brand images and SEO metadata are editable for storefront merchandising.	P1
FR-M04-11	Category serviceability restrictions can be set — for example, a category not shippable to certain PIN codes.	P2

M04.3 Business rules

- A product inherits licence requirements from every ancestor category, not just its immediate parent.
- Brand approval is independent of product approval — an approved brand does not imply an approved listing.
- Moving a category re-evaluates the commission resolution for future orders only; existing snapshots are untouched.

MODULE 05 · PHASE 1A · DAYS 11–30

Product Management

Five product types · variants · bulk import · three-stage approval · supplier sync

M05.1 Purpose

Everything a seller needs to get a catalog onto the platform, and everything Admin needs to control what actually goes live.

M05.2 Product types

Type	Definition
Simple	One SKU, one price, no options.
Variable	Parent product with variants across options — size, colour, capacity. Each variant is its own SKU with its own stock and price.
Bulk	Sold in large quantities with quantity-break pricing; typically B2B.
Wholesale	Restricted to buyers holding a wholesale-eligible role; MOQ enforced.
Pack Size	Same item sold in defined pack configurations, each with its own SKU and price.

M05.3 Mandatory product attributes

Every product, regardless of type, carries: **SKU**, **barcode**, **MOQ**, **GST rate** and **HSN code**. These are not optional fields — invoicing and tax reporting depend on them, and a product missing any of them cannot pass approval.

M05.4 Approval chain

Category Approval → Brand Approval → Product Approval → LIVE

Each gate is independent. A product cannot skip a gate, and failing any gate returns the product to draft with a stated reason.

M05.5 Functional requirements

ID	Requirement	Pri
FR-M05-01	Sellers create products of all five types with full CRUD on their own catalog.	P0
FR-M05-02	Variable products support multiple option axes with a generated variant matrix, each variant carrying its own SKU, price, stock and images.	P0
FR-M05-03	SKU, barcode, HSN, GST rate and MOQ are mandatory and validated before submission for approval.	P0
FR-M05-04	SKU is unique within a seller's catalog.	P0
FR-M05-05	Products support multiple images with ordering, a primary image, and per-variant imagery.	P0
FR-M05-06	CSV and Excel import creates and updates products in bulk.	P0
FR-M05-07	Import validates every row and produces a downloadable error report identifying the row, column and reason for each failure. Valid rows are imported; invalid rows are not.	P0
FR-M05-08	Import is previewable before commit, showing counts of creates, updates and rejections.	P1
FR-M05-09	Products pass the three-stage approval chain before going live.	P0
FR-M05-10	Admin approves or rejects a product with a reason; rejection returns it to draft and notifies the seller.	P0
FR-M05-11	Editing a live product's material attributes — title, images, category, HSN, GST — returns it to pending approval. Price and stock edits do not.	P0
FR-M05-12	Products carry a source attribute — <code>manual</code> , <code>csv</code> or <code>api</code> — and API-sourced products carry a supplier reference.	P0
FR-M05-13	Supplier adapters sync stock, price, image and description for API-sourced products. Capped at two supplier integrations.	P0
FR-M05-14	A seller cannot list under a category whose licence requirements it does not satisfy.	P0
FR-M05-15	Admin can bulk-approve, bulk-reject, bulk-activate and bulk-deactivate products from a filtered list.	P1
FR-M05-16	Admin can list and manage Own Stock products, distinguished by ownership from seller products.	P0
FR-M05-17	Products carry Legal Metrology declarations where the category requires them.	P1
FR-M05-18	Prohibited-category listing attempts are blocked and logged.	P0

M05.6 Edge cases

Case	Handling
Supplier sync changes price on a product in someone's cart	Cart holds the tier-snapshotted price; the buyer is shown a price-change notice at checkout and must confirm before paying.
Supplier sync drops stock to zero mid-order	Order acceptance fails at the vendor step; the sub-order is cancelled and refunded. This is the main dropshipping failure mode and must be visible in reporting.
Import file with 50,000 rows	Processed asynchronously with progress reporting; the user is not blocked and receives a completion notification.
Seller edits GST rate on a product with historical orders	Allowed for future orders. Historical invoices retain their snapshotted rate.
Variant deleted while in an open cart	Cart line is marked unavailable and excluded from checkout with a clear message.

MODULE 06 · PHASE 1A–1B

Inventory Management

Owner buckets · reservation · low-stock alerts · own-stock separation

M06.1 Purpose

Tracks available quantity per product or variant, separated by owner bucket so that platform inventory is reportable independently of vendor inventory — a stated requirement of Model C.

M06.2 The reservation model

Stock has two numbers: **quantity on hand** and **reserved quantity**. Available stock is the difference. Reservation happens at payment confirmation, not when an item enters a cart — reserving at cart-add invites abandoned carts to starve real buyers, and reserving at dispatch invites overselling.

on_hand	physical or seller-declared quantity
reserved	allocated to confirmed, unshipped sub-orders
available	on_hand - reserved ← what the storefront shows
reserve	at Payment Verified
release	on cancellation, vendor rejection, or RTO completion
decrement	on Shipped (on_hand and reserved both fall)

M06.3 Functional requirements

ID	Requirement	Pri
FR-M06-01	Inventory is tracked per product or variant with an explicit owner bucket of <code>vendor</code> or <code>own_stock</code> .	P0
FR-M06-02	Available quantity is derived as on-hand minus reserved and is what the storefront displays.	P0
FR-M06-03	Stock is reserved at payment confirmation and released on cancellation, vendor rejection or RTO completion.	P0
FR-M06-04	Concurrent checkout on the last unit cannot oversell; reservation is atomic.	P0
FR-M06-05	Sellers update stock individually and in bulk by CSV.	P0

ID	Requirement	Pri
FR-M06-06	Low-stock alerts fire at a per-product configurable threshold, to the seller and to Admin for own stock.	P0
FR-M06-07	Out-of-stock products remain visible but are not purchasable, unless the seller has hidden them.	P1
FR-M06-08	Every stock movement is recorded with reason, actor, delta and resulting balance.	P0
FR-M06-09	Own-stock inventory, orders, shipping and sales are reportable separately from vendor stock.	P0
FR-M06-10	Supplier-synced inventory updates on the adapter's schedule and records the sync timestamp.	P0
FR-M06-11	Admin can adjust own-stock quantities with a mandatory reason, producing an audit entry.	P0
FR-M06-12	Multi-warehouse inventory for own stock.	P2

Dropshipping caveat. For Model A products, our stock figure is the supplier's last reported number, not an observed truth. Between syncs it can be wrong. This is why vendor rejection is a first-class order state rather than an error condition, and why the sub-order cancellation path must be as reliable as the happy path.

MODULE 07 · PHASE 1B · DAYS 25–45

Pricing and Commission Engine

Tiered pricing by buyer role · MOQ · discount rules · commission resolution · margin override

M07.1 Purpose

The commercial core. Decides what price a given buyer sees for a given product, and what commission the platform earns on the resulting sale. Both are centralised and fully Admin-configurable, and both are **snapshotted at order time** so history stays reproducible.

M07.2 Price resolution by buyer role

A single product may be sold at different prices to different buyer types. Price tiers are records attached to a product, each naming a buyer role, a price and a minimum quantity.

Tier	Applies to	MOQ behaviour
Retail (B2C)	Retail Customer	No MOQ. Default fallback for any buyer with no eligible tier.
Wholesale	Wholesaler, B2B Buyer	MOQ enforced at the cart line.
Dealer	Dealer	MOQ enforced.
Distributor	Distributor	MOQ enforced.

```

resolve_price(product, buyer):
    tiers ← price_tiers where product = product AND buyer_role ∈ buyer.capabilities
    if tiers is empty          → retail price
    if quantity < tier.min_qty  → block the line, state the MOQ
    if multiple tiers eligible → lowest price wins, MOQ satisfied
    → snapshot (price, tier_id) onto the cart line

```

M07.3 Commission resolution

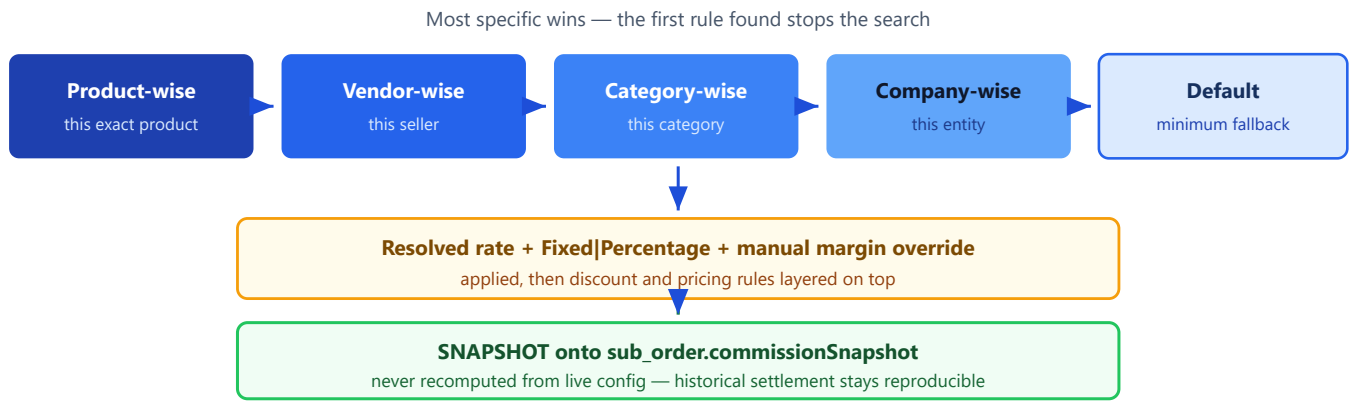


Figure M07.1 — Commission resolution order and snapshotting.

M07.4 Functional requirements

ID	Requirement	Pri
FR-M07-01	Commission rules are scoped to product, vendor, category, company or default, and resolve most-specific-first.	P0
FR-M07-02	Commission type is Fixed or Percentage.	P0
FR-M07-03	A minimum default rule always exists so resolution can never return nothing.	P0
FR-M07-04	Automatic margin calculation is provided with a manual override capability, and overrides are audit-logged with a reason.	P0
FR-M07-05	The resolved commission is snapshotted onto the sub-order at order placement and never recomputed.	P0
FR-M07-06	Multiple price levels — retail, wholesale, dealer, distributor — resolve by buyer role.	P0
FR-M07-07	MOQ is enforced on B2B tiers at the cart line, with a clear message stating the required quantity.	P0
FR-M07-08	Discount rules and pricing rules layer on top of the resolved tier price with a defined precedence.	P0
FR-M07-09	Admin can simulate pricing and commission for a hypothetical product, buyer role and quantity without placing an order.	P1
FR-M07-10	Commission rate changes take effect only for orders placed after the change; a change never alters an existing snapshot.	P0
FR-M07-11	Sellers can view the commission rates applicable to them in the Seller Panel.	P0
FR-M07-12	Commission rule changes are audit-logged with before and after values.	P0
FR-M07-13	Own Stock products carry no commission — the platform is the seller.	P0

Never recompute historical financial values. Price, commission and GST rate are snapshotted onto the order. Recomputing them from live configuration at settlement or reporting time would make yesterday's settlement statement change when today's configuration changes — which is indefensible in a vendor dispute and unauditable at year end.

Search and Discovery

Full-text search · faceted filters · sorting · category browse · recommendations

Blocking decision — see §44 D3. MongoDB regular-expression search will not scale to a 100,000-SKU marketplace catalog with facets. The choice between Atlas Search and Elasticsearch must be made *before* catalog work begins, because it determines how products are indexed, not merely how they are queried. Deferring this decision forces a rewrite of the catalog read path later.

M08.1 Functional requirements

ID	Requirement	Pri
FR-M08-01	Full-text search across product title, description, brand, category and SKU.	P0
FR-M08-02	Search tolerates minor misspellings and supports partial-word matching.	P1
FR-M08-03	Faceted filtering by category, brand, price range, rating, availability and seller.	P0
FR-M08-04	Facet counts reflect the currently applied filter set.	P1
FR-M08-05	Sorting by relevance, price ascending and descending, newest, and rating.	P0
FR-M08-06	Prices shown in results reflect the viewing buyer's tier.	P0
FR-M08-07	Only approved, active, in-catalog products from active sellers appear in results.	P0
FR-M08-08	Search-as-you-type suggestions for products, categories and brands.	P1
FR-M08-09	Search results paginate and return within the p95 latency target at full catalog volume.	P0
FR-M08-10	Zero-result searches are logged for merchandising analysis.	P1
FR-M08-11	Related and recently viewed products on the product detail page.	P1
FR-M08-12	Personalised recommendations from purchase history.	P2

Cart and Wishlist

Persistent cart · tier snapshot · MOQ validation · wishlist · save for later

M09.1 Functional requirements

ID	Requirement	Pri
FR-M09-01	Cart persists per user across devices and sessions.	P0
FR-M09-02	Each cart line stores the resolved price and the tier that produced it.	P0
FR-M09-03	The cart can hold items from multiple sellers and multiple business models simultaneously.	P0
FR-M09-04	Quantity changes re-validate MOQ and available stock, and re-resolve the tier if a quantity break is crossed.	P0
FR-M09-05	The cart shows a per-seller grouping so the buyer understands the split before checkout.	P0

ID	Requirement	Pri
FR-M09-06	Unavailable lines — delisted, out of stock, seller suspended — are flagged and excluded from checkout with an explanation.	P0
FR-M09-07	A price change since the item was added is surfaced explicitly and requires acknowledgement before payment.	P0
FR-M09-08	Wishlist add, remove, and move-to-cart.	P0
FR-M09-09	Save-for-later moves a line out of the active cart without losing it.	P1
FR-M09-10	A guest cart merges into the user cart on login without losing lines.	P1
FR-M09-11	Cart totals show item subtotal, discounts, shipping and tax as separate lines.	P0
FR-M09-12	Serviceability is checked against the delivery PIN code before payment, not after.	P0

MODULE 10 · PHASE 1B · DAYS 25–45

Checkout and Order Splitting

Address selection · re-validation · multi-vendor split · order formation

M10.1 Purpose

Converts a validated cart into a parent Order and its per-seller Sub-Orders. This is the point at which every commercial value is fixed, and the point at which the platform's multi-vendor nature becomes concrete.

M10.2 Checkout and split flow

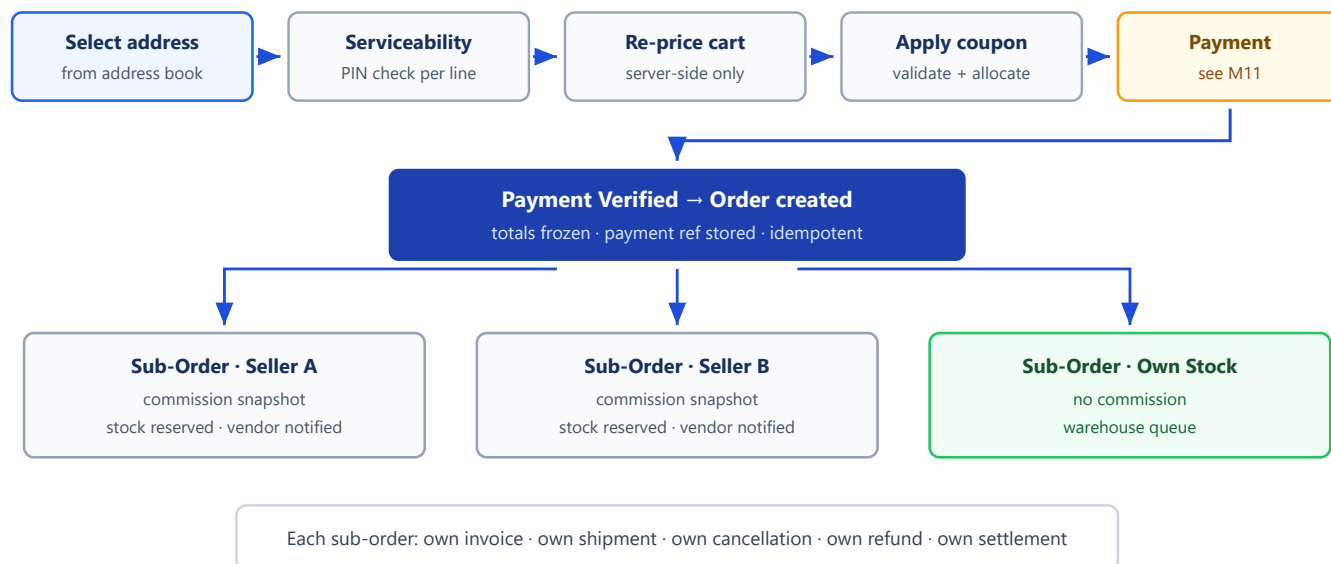


Figure M10.1 — Checkout to split. Everything commercial is frozen at the moment the order is created.

M10.3 Functional requirements

ID	Requirement	Pri
FR-M10-01	Buyers maintain an address book with multiple saved addresses and a default.	P0
FR-M10-02	Checkout re-validates stock, price, MOQ, tier eligibility, coupon validity and PIN serviceability server-side before payment is initiated.	P0

ID	Requirement	Pri
FR-M10-03	All amounts are computed server-side; no client-supplied price or total is trusted.	P0
FR-M10-04	On payment confirmation the cart splits into one sub-order per seller, plus one for own stock if present.	P0
FR-M10-05	Each sub-order carries its own status, commission snapshot, invoice reference and settlement record.	P0
FR-M10-06	Shipping charges are computed and allocated per sub-order.	P0
FR-M10-07	Coupon value is allocated proportionally across affected sub-orders so refunds stay arithmetically correct.	P0
FR-M10-08	Order placement is idempotent; a duplicate submission does not create a second order.	P0
FR-M10-09	B2B buyers supply a GSTIN before checkout; it cannot be added to an invoice afterwards.	P0
FR-M10-10	The buyer sees the split and estimated delivery per sub-order before paying.	P0
FR-M10-11	Stock is reserved atomically at payment confirmation.	P0
FR-M10-12	Anti-fraud checks run at placement — velocity, address anomalies, prior chargebacks.	P0

MODULE 11 · PHASE 1B · DAYS 25–45

Payment

Razorpay checkout · signature verification · webhooks · refunds · no COD

Online payments only. No Cash on Delivery. COD is an explicit exclusion. Every order is paid in full before processing. No COD collection, reconciliation or agent-cash handling exists anywhere in the system.

M11.1 Payment flow

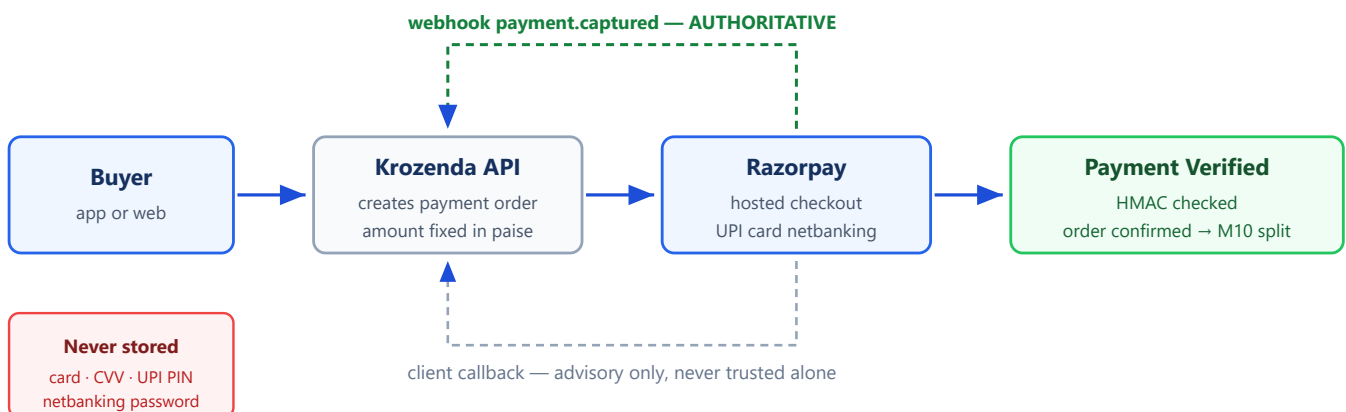


Figure M11.1 — Payment capture. The server-to-server webhook confirms the order, not the browser.

M11.2 Functional requirements

ID	Requirement	Pri
FR-M11-01	Payment methods supported: UPI, credit card, debit card, net banking, wallet.	P0
FR-M11-02	No Cash on Delivery anywhere in the system.	P0

ID	Requirement	Pri
FR-M11-03	Payment credentials are captured on the gateway's hosted interface; the platform never receives, sees or stores them.	P0
FR-M11-04	The payment order is created server-side with the amount in paise and the internal order reference as receipt.	P0
FR-M11-05	The checkout callback signature is HMAC-verified before any state change.	P0
FR-M11-06	Order confirmation is driven by the verified <code>payment.captured</code> webhook, not by the client callback.	P0
FR-M11-07	Every webhook is signature-verified against the webhook secret; unsigned or mismatched payloads are rejected and logged as security events.	P0
FR-M11-08	Webhook handlers are idempotent; repeated delivery of the same event changes nothing.	P0
FR-M11-09	Full and partial refunds are supported through the gateway's Refunds API.	P0
FR-M11-10	A refund on a vendor sub-order also reverses the corresponding Route transfer; the two legs are tracked as one operation.	P0
FR-M11-11	Payment failure leaves the cart intact and allows retry without re-entering details.	P0
FR-M11-12	A nightly reconciliation compares gateway records against platform records and reports every discrepancy.	P0
FR-M11-13	Orphan payments — money captured with no order created — are detected and auto-refunded.	P0
FR-M11-14	Duplicate payments against one order reference are detected and the surplus refunded.	P0
FR-M11-15	Payment records store only reference, amount, method, status and timestamp.	P0

Order Management

Lifecycle state machine · vendor assignment · accept/reject · cancellation · order views

M12.1 Order lifecycle

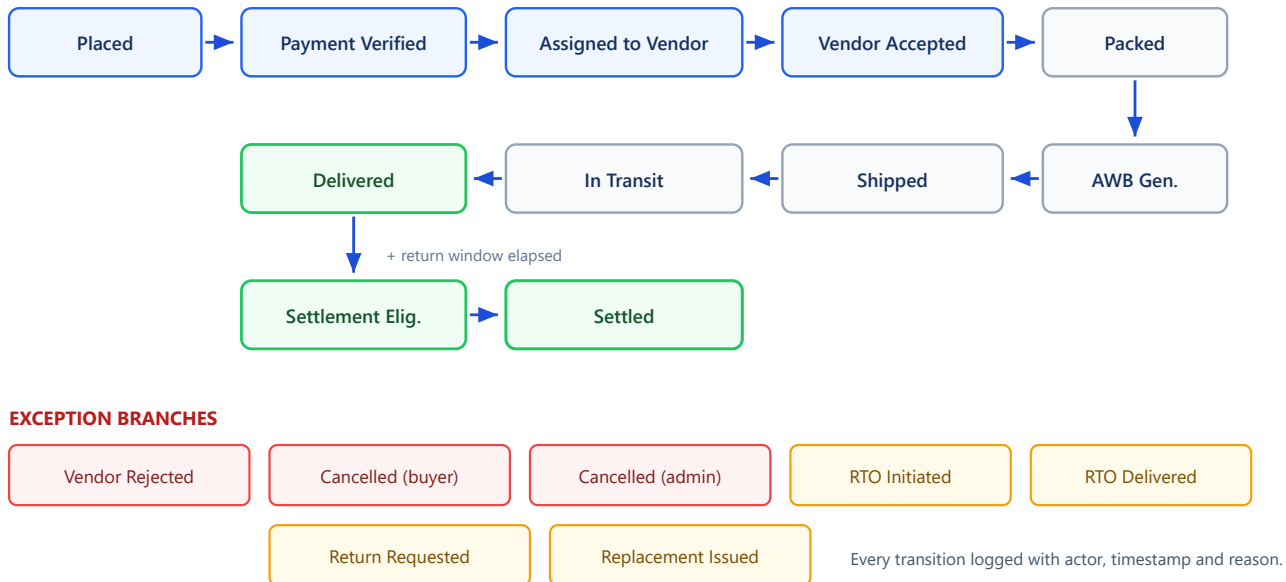


Figure M12.1 — Sub-order lifecycle. Each sub-order moves independently of its siblings.

M12.2 Functional requirements

ID	Requirement	Pri
FR-M12-01	The sub-order state machine implements every state and exception branch in Figure M12.1, and rejects invalid transitions.	P0
FR-M12-02	Every transition is logged with actor, timestamp and reason.	P0
FR-M12-03	Sub-orders are auto-assigned to the owning vendor on payment confirmation and the vendor is notified.	P0
FR-M12-04	Dropshipping orders are auto-forwarded to the supplier by adapter where an integration exists, and by notification otherwise.	P0
FR-M12-05	Vendors accept or reject an assigned sub-order within a configurable window; rejection requires a reason.	P0
FR-M12-06	Vendor rejection cancels that sub-order, refunds the buyer proportionally, reverses the transfer and releases reserved stock.	P0
FR-M12-07	Buyers cancel a sub-order free of charge at any time before AWB generation.	P0
FR-M12-08	Cancelling one sub-order never affects its siblings.	P0
FR-M12-09	Admin can cancel any sub-order with a mandatory reason, at any pre-delivery state.	P0
FR-M12-10	Buyers see a unified order view with per-sub-order status, tracking and expected delivery.	P0
FR-M12-11	Sellers see only their own sub-orders, with filters by status, date and fulfilment urgency.	P0

ID	Requirement	Pri
FR-M12-12	Admin sees all orders and sub-orders with filters across buyer, seller, model, status, date and value.	P0
FR-M12-13	Packing slips are generated per sub-order and printable individually or in a batch.	P0
FR-M12-14	Orders stalled beyond configurable thresholds appear in an Admin exception queue.	P0
FR-M12-15	Buyers can reorder a past order, subject to current price, stock and eligibility.	P1
FR-M12-16	Order status changes notify the buyer by push, SMS and email per the notification matrix.	P0

M12.3 Edge cases

Case	Handling
Vendor never responds to assignment	Auto-reject on timeout, cancel and refund that sub-order, flag the vendor's response-rate metric.
One sub-order cancelled, coupon applied across all	Coupon value was allocated proportionally at M10.07, so the refund is arithmetically correct without re-deriving the discount.
Buyer cancels between packing and AWB	Allowed — the gate is AWB generation, not packing. The vendor is notified to unpack.
Delivery webhook never arrives	Sub-order stays pre-Delivered, settlement stays gated, and it surfaces in the stale-order report. Fails closed.
Seller suspended with sub-orders in flight	Existing sub-orders continue to fulfilment; no new assignments. Settlement releases normally unless the suspension is fraud-related.

MODULE 13 · PHASE 1C · DAYS 40–62

Fulfilment and Shipping

Shiprocket · AWB · courier allocation · labels · pickup · tracking webhooks

M13.1 Fulfilment flow



Figure M13.1 — Fulfilment. Delivery confirmation comes from the courier, never from the vendor's assertion.

M13.2 Functional requirements

ID	Requirement	Pri
FR-M13-01	AWB generation through Shiprocket per sub-order, using the vendor's registered pickup address or the platform warehouse for own stock.	P0

ID	Requirement	Pri
FR-M13-02	Courier allocation via Shiprocket, with the selected courier stored on the shipment.	P0
FR-M13-03	Shipping labels are generated and downloadable individually and in batch.	P0
FR-M13-04	Pickup scheduling is requested through Shiprocket and its status is surfaced to the vendor.	P0
FR-M13-05	Tracking events arrive by webhook and are stored as an ordered event list on the shipment.	P0
FR-M13-06	Buyers see live tracking with the courier name, AWB and event history.	P0
FR-M13-07	Delivery confirmation transitions the sub-order to Delivered and starts the return window clock.	P0
FR-M13-08	Vendors record package weight and dimensions before AWB generation; these feed courier rating.	P0
FR-M13-09	PIN-code serviceability is checked at checkout, not after payment.	P0
FR-M13-10	Shipping charges are computed per sub-order and allocated to buyer or vendor per configuration.	P0
FR-M13-11	A failed AWB request surfaces to the vendor with the provider's reason and is retryable.	P0
FR-M13-12	Own-stock shipments are distinguishable from vendor shipments in all reporting.	P0
FR-M13-13	Multi-package shipments for a single sub-order.	P2

MODULE 14 · PHASE 1D · DAYS 55–72

Returns, Replacement and RTO

No-Return default · three exceptions · photo evidence · admin adjudication · RTO paths

M14.1 The policy

The platform's default is **No Return**. Returns are accepted only for three reasons, each requiring photographic evidence and Admin approval.

Accepted reason	Definition
Damaged Product	Arrived broken, defective or damaged in transit.
Wrong Product	Item delivered is not the item ordered — different product, variant, size or colour.
Missing Product	An item shown as delivered is absent from the package.

Unresolved conflict — client decision required (\$44 D5). A blanket No-Return default must be reconciled with the Consumer Protection (E-Commerce) Rules, 2020, which require a marketplace entity to publish clear return, refund and exchange terms and prohibit unfair trade practices. This is raised rather than silently resolved, per the project's working conventions. The client's legal counsel must confirm the position is defensible for every category listed, and whether a cooling-off window is required. **The build implements whatever the client decides; it does not choose.**

M14.2 Return flow

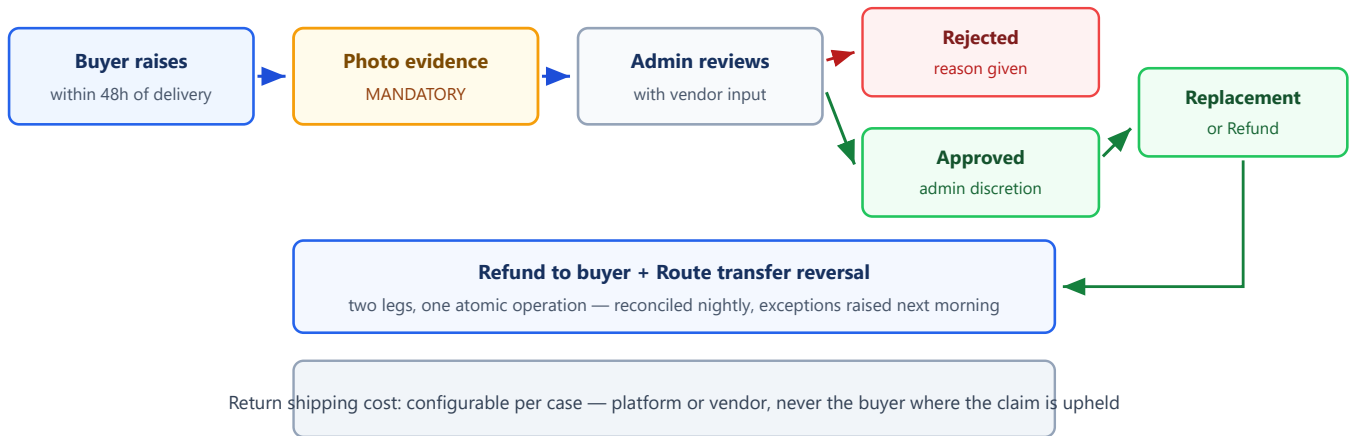


Figure M14.1 — Return adjudication. Evidence is mandatory; the refund and the transfer reversal move together.

M14.3 Functional requirements — returns

ID	Requirement	Pri
FR-M14-01	Returns are accepted only for Damaged, Wrong or Missing product. No other reason is selectable.	P0
FR-M14-02	A return request must be raised within a configurable window, default [48] hours from delivery.	P0
FR-M14-03	Photographic evidence is mandatory; a request cannot be submitted without at least one image.	P0
FR-M14-04	Admin approves or rejects each request with a stated reason, consulting the vendor.	P0
FR-M14-05	An approved request resolves to a replacement or a refund at Admin discretion, having regard to stock and buyer preference.	P0
FR-M14-06	Return shipping cost allocation is configurable per case between platform and vendor.	P0
FR-M14-07	Refunds return to the original payment instrument; no cash refunds and no refunds to a different account.	P0
FR-M14-08	An approved refund reverses the corresponding Route transfer; both legs are tracked and reconciled.	P0
FR-M14-09	Where a transfer was already released and the vendor's balance is insufficient, the shortfall becomes a recoverable deducted from the next settlement cycle.	P0
FR-M14-10	Partial returns of a multi-item sub-order refund only the affected items and reverse only that proportion.	P0
FR-M14-11	An open return claim blocks settlement release for that sub-order.	P0
FR-M14-12	Buyers and vendors see return status and history on their own surfaces.	P0
FR-M14-13	Return reasons and outcomes are reportable by vendor, category and product to surface quality problems.	P1

M14.4 RTO handling

The No-Return policy does **not** cover Return to Origin. RTO has its own paths and must be built explicitly.

ID	Requirement	Pri
FR-M14-14	RTO is triggered by undelivered shipment, address failure, or customer unreachable, detected from courier tracking events.	P0

ID	Requirement	Pri
FR-M14-15	Return shipping cost on RTO is configurable between platform and vendor.	P0
FR-M14-16	On RTO completion the vendor's settlement for that sub-order is reversed.	P0
FR-M14-17	On RTO completion, inventory is restored for Own Stock and Marketplace models.	P0
FR-M14-18	Where the RTO is not buyer-attributable, the buyer is refunded in full including shipping.	P0
FR-M14-19	Where the RTO is buyer-attributable, the product value is refunded and freight actually incurred may be deducted, disclosed before deduction.	P0
FR-M14-20	Repeated buyer-attributable RTOs flag the account for review and possible restriction.	P1

MODULE 15 · PHASE 1D · DAYS 55–72

Settlement and Payouts

Route transfers · holds · release gate · settlement ledger · reversal · RazorpayX payouts

M15.1 Purpose

Moves money to vendors correctly, automatically and compliantly, and provides the ledger that makes every rupee explicable. This module carries the platform's highest financial and regulatory risk.

M15.2 Settlement flow

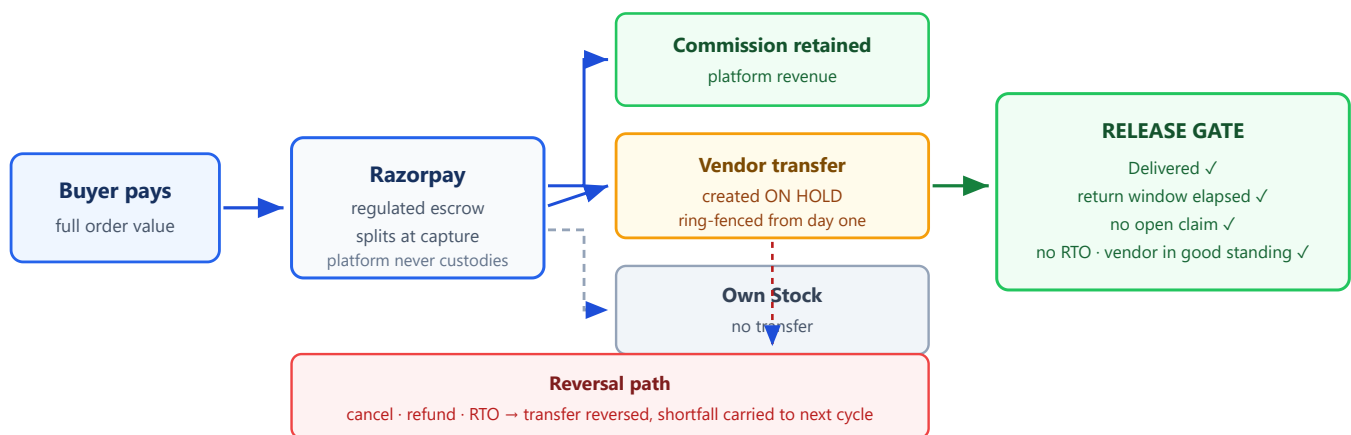


Figure M15.1 — Route settlement with hold and release gate.

M15.3 Functional requirements — vendor settlement

ID	Requirement	Pri
FR-M15-01	Vendor settlement is implemented as Razorpay Route transfers, holds and reversals — not a home-grown payout ledger.	P0
FR-M15-02	Every vendor must hold a Route linked account with completed KYC before any transfer is possible.	P0
FR-M15-03	A vendor's share is transferred on hold at payment capture, ring-fencing it against that payment.	P0
FR-M15-04	A held transfer releases only when all five release-gate conditions are satisfied.	P0
FR-M15-05	Settlement cycles are configurable per vendor: automatic, manual, weekly or monthly.	P0

ID	Requirement	Pri
FR-M15-06	A scheduled job evaluates eligibility and releases due transfers without human intervention.	P0
FR-M15-07	Every settlement writes an immutable record: gross, commission, each deduction, net, transfer id, UTR, status and timestamp.	P0
FR-M15-08	Deductions supported: commission, gateway charge, shipping borne by vendor, RTO and return costs, penalties, prior-cycle recoveries, and tax withheld.	P0
FR-M15-09	Vendors see a settlement statement per sub-order and per cycle, with the full arithmetic.	P0
FR-M15-10	Vendors see pending settlements with the expected release date.	P0
FR-M15-11	Admin can place a manual hold on a vendor's settlement with a reason, and release it.	P0
FR-M15-12	Transfer reversal is supported for cancellation, refund and RTO, including partial reversal.	P0
FR-M15-13	Where a reversal exceeds the vendor's available balance, the shortfall is recorded as recoverable and deducted from subsequent cycles until cleared.	P0
FR-M15-14	Settlement webhooks — <code>transfer.processed</code> , <code>transfer.failed</code> — update status; a failure blocks further releases to that vendor and alerts Admin.	P0
FR-M15-15	A nightly reconciliation independently fetches gateway records and reports any drift.	P0
FR-M15-16	Settlement reports are exportable by vendor, period and status.	P0

M15.4 Settlement arithmetic

```
gross sale value
- platform commission      (snapshotted at order time – never recomputed)
- payment gateway charge   (where borne by the vendor)
- shipping charge          (where borne by the vendor)
- RTO / return shipping    (per case allocation)
- penalties                (SLA breaches, per Vendor Agreement)
- prior-cycle recoveries   (shortfalls carried forward)
- TCS (GST s.52) · TDS (IT s.194-0)
= NET SETTLED → transfer id · UTR · status · timestamp
```

M15.5 Automated payouts — RazorpayX ADD

Two rails, and they are not interchangeable. Vendor sale proceeds move on **Route** (M15.3). Krozenda's own money — salaries, reimbursements, incentives, goodwill payments outside the gateway refund window — moves on **RazorpayX Payouts**. Paying vendors by Payout API would mean collecting customer money into Krozenda's account and disbursing it, which requires a Payment Aggregator licence Krozenda does not hold. This boundary is a licensing constraint, not an implementation preference.

Money movement	Rail	Why
Vendor sale proceeds	Route	Third-party money; must split at source
Commission, own-stock revenue	Route (retained)	Settles to platform account normally
Customer refund	Refunds API	Reverses the original payment; not a payout
Staff salary	Payouts	Platform's own money to its own employees

Money movement	Rail	Why
Employee reimbursement	Payouts	Same
Vendor incentive paid by platform	Payouts	Platform's own money, a business expense — distinct from sale proceeds
Goodwill payment outside refund window	Payouts	The original payment can no longer be reversed

ID	Requirement	Pri
FR-M15-17	The system maintains contacts and fund accounts for payout beneficiaries, created once and reused.	ADD
FR-M15-18	A payout run reads an approved register, validates every row, checks the total against available balance, and halts rather than paying partially.	ADD
FR-M15-19	Maker-checker: a run is prepared by one user and released by another. No single account can both prepare and release.	ADD
FR-M15-20	Every payout carries a deterministic idempotency key derived from period and beneficiary — never a random value.	ADD
FR-M15-21	Payout mode is selectable per beneficiary: IMPS, NEFT, RTGS or UPI.	ADD
FR-M15-22	Payout status is tracked per beneficiary; <i>failed</i> and <i>reversed</i> are surfaced distinctly, never collapsed into a generic error.	ADD
FR-M15-23	Low balance queues payouts rather than failing them, and alerts finance immediately.	ADD
FR-M15-24	A run report lists paid, failed, reversed and pending with UTRs, retained for audit.	ADD
FR-M15-25	Payout initiation and approval are distinct privileged permissions held by a small number of finance roles.	ADD

Payroll scope boundary. M15.5 specifies *disbursement* — moving an approved net-payable figure reliably. Statutory payroll computation (PF, ESI, professional tax, TDS on salary under s.192, leave and attendance, Form 16) is **not included** and would be a separate engagement, or handled in dedicated payroll software that supplies the net figures to this module.

MODULE 16 · SCOPE ADDITION — NOT IN THE SOW

Wallet and Store Credit

Refund credit · wallet ledger · redemption at checkout · regulatory constraints

This module is a scope addition (ADD). A platform wallet is not in the agreed Phase 1 scope and does not appear in the Scope of Work — "wallet" there refers only to third-party wallets as a payment method at checkout. It is specified here in full because the client requested it. **It requires a written change request covering cost and timeline before any work begins**, and it must not be assumed to be inside the 90-day commitment.

M16.1 Purpose

A wallet holds a balance the buyer can redeem against future purchases. It reduces refund friction — credit is instant where a gateway refund takes days — and it retains value on the platform instead of returning it to the buyer's bank.

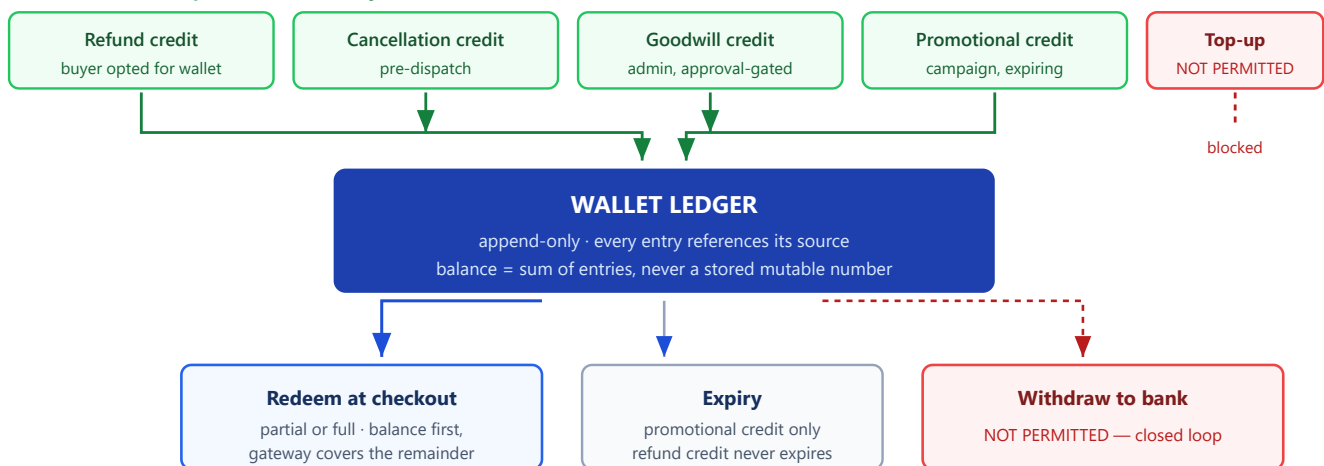
M16.2 The regulatory constraint that shapes the design

A reloadable wallet is a Prepaid Payment Instrument. If customers can top up a wallet with fresh money, the platform is issuing a PPI, which requires RBI authorisation, an escrow account and its own compliance regime. That is a substantially larger undertaking than a commerce feature and is **not** proposed here.

What is specified below is a **closed-loop store credit wallet**: it is funded only by refunds, cancellations, goodwill credits and promotional credits issued by the platform. It is **non-reloadable** — a customer cannot add their own money — and **non-withdrawable** — the balance cannot be cashed out to a bank account. It can only be spent on the platform. This is the variant that avoids the PPI question, and the client's legal counsel should confirm the position before launch (\$44 D8).

M16.3 Wallet flow

CREDIT SOURCES — platform-issued only



Closed loop: platform-funded in, platform-spend out. No cash in, no cash out.

Figure M16.1 — Closed-loop store credit wallet. Top-up and withdrawal are deliberately absent.

M16.4 Functional requirements

ID	Requirement	Pri
FR-M16-01	Each buyer has one wallet, denominated in INR, implemented as an append-only ledger.	ADD
FR-M16-02	Balance is derived by summing ledger entries; it is never stored as an independently mutable number.	ADD
FR-M16-03	Every ledger entry records type, amount, source reference (sub-order, campaign, admin action), actor, timestamp and resulting balance.	ADD
FR-M16-04	Credit sources are limited to refund, cancellation, goodwill and promotional credit. Customer top-up is not supported.	ADD
FR-M16-05	Withdrawal to a bank account is not supported. The wallet is closed-loop.	ADD
FR-M16-06	On an approved refund the buyer chooses between original payment method and wallet credit; wallet credit is instant, and this is stated at the point of choice.	ADD
FR-M16-07	Wallet balance can be redeemed at checkout in part or full; any shortfall is charged to the gateway in the same transaction.	ADD
FR-M16-08	Wallet redemption is allocated proportionally across sub-orders so refunds remain arithmetically correct.	ADD

ID	Requirement	Pri
FR-M16-09	A refund on an order partly paid by wallet returns the wallet portion to the wallet and the gateway portion to the instrument.	ADD
FR-M16-10	Promotional credit may carry an expiry; refund and cancellation credit never expires.	ADD
FR-M16-11	Buyers see full wallet history with running balance and the reason for every entry.	ADD
FR-M16-12	Admin can issue goodwill credit subject to an approval threshold, with a mandatory reason, audit-logged.	ADD
FR-M16-13	Wallet liability — total outstanding balance across all users — is reportable for accounting.	ADD
FR-M16-14	Wallet balance cannot go negative under any code path; concurrent redemption is atomic.	ADD
FR-M16-15	Wallet credit does not affect vendor settlement: the vendor is paid from the original captured payment regardless of how the buyer funded it.	ADD
FR-M16-16	Account suspension freezes the wallet; balance is preserved and visible but not redeemable.	ADD

M16.5 The settlement subtlety

Wallet redemption is a discount to the platform, not to the vendor. If a buyer pays ₹1,000 of which ₹200 comes from wallet credit, only ₹800 reaches the gateway — but the vendor is still owed its full share of the ₹1,000 sale. The ₹200 must be funded by the platform from its own money. This is why wallet is not a trivial feature bolted onto checkout: it changes the arithmetic of the Route split, and the platform must top up the difference so the vendor is made whole. Any implementation that quietly short-pays the vendor by the wallet amount is a serious defect and a breach of the Vendor Agreement.

M16.6 Edge cases

Case	Handling
Full order value covered by wallet	No gateway transaction occurs for the buyer, but the vendor's Route transfer must still be funded by the platform. Requires a funded platform float.
Order cancelled that was partly wallet-paid	Wallet portion returns to wallet; gateway portion refunds to instrument. Both legs reconciled.
Promotional credit expires mid-checkout	Balance is re-derived at payment time; an expired entry no longer counts and the buyer is told before paying.
Concurrent redemption on two devices	Atomic ledger append with balance check prevents double-spend.
Refund exceeds remaining wallet balance	Impossible by construction — refund credits, never debits.

Coupons, Offers and Discounts

Coupon codes · automatic offers · pricing rules · allocation across sub-orders

M17.1 Functional requirements

ID	Requirement	Pri
FR-M17-01	Admin creates coupons with a code, discount type (fixed or percentage), value, and an optional maximum discount cap.	P0
FR-M17-02	Coupons carry validity windows, total usage caps and per-user usage caps.	P0
FR-M17-03	Coupons can be scoped to categories, products, sellers, buyer roles or minimum order value.	P0
FR-M17-04	Coupon validity is re-checked server-side at payment time, not only when applied.	P0
FR-M17-05	Discount value is allocated proportionally across the sub-orders it applies to.	P0
FR-M17-06	Who bears a discount — platform or vendor — is configurable per coupon and flows into settlement.	P0
FR-M17-07	Automatic offers apply without a code where their conditions are met.	P1
FR-M17-08	Coupon stacking rules are explicit; by default coupons do not stack.	P0
FR-M17-09	Coupon usage is reportable — redemptions, discount given, orders influenced.	P1
FR-M17-10	Expired or exhausted coupons fail with a clear reason rather than a generic error.	P0

Reviews and Ratings

Verified-purchase reviews · photo uploads · moderation · aggregate ratings

M18.1 Functional requirements

ID	Requirement	Pri
FR-M18-01	Only a buyer with a delivered sub-order containing the product may review it.	P0
FR-M18-02	A review carries a star rating from 1 to 5 and optional text.	P0
FR-M18-03	Buyers may attach photographs to a review.	P0
FR-M18-04	One review per buyer per product, editable within a configurable window.	P0
FR-M18-05	Aggregate rating and rating distribution are shown on the product page and recalculated on each change.	P0
FR-M18-06	Admin can moderate — approve, reject or remove — any review, with a reason recorded.	P0
FR-M18-07	Reviews can be reported by other users, entering a moderation queue.	P1
FR-M18-08	Sellers may respond publicly once per review.	P1
FR-M18-09	Reviews are marked "Verified Purchase" — all reviews qualify by construction under FR-M18-01.	P0
FR-M18-10	On account deletion, reviews are anonymised rather than removed, to preserve product ratings.	P1
FR-M18-11	Suspicious review patterns — volume, timing, repetition — are flagged for moderation.	P2

Notification Engine

FCM push · SMS India Hub · SMTP email · templates as data · DLT compliance

M19.1 Purpose

One engine, three channels, templates held as configurable records. Because DLT-approved SMS text cannot be changed freely once registered, template content must never be embedded in code.

M19.2 Channel matrix

Event	Push	SMS	Email	Recipient
OTP — registration / login	—	Yes	—	User
Email verification	—	—	Yes	User
Order placed	Yes	Yes	Yes	Buyer
New order assigned	Yes	Yes	Yes	Seller
Vendor accepted / rejected	Yes	—	Yes	Buyer
Shipped + AWB	Yes	Yes	Yes	Buyer
Out for delivery	Yes	Yes	—	Buyer
Delivered	Yes	Yes	Yes	Buyer
Cancellation confirmed	Yes	Yes	Yes	Buyer
Return approved / rejected	Yes	Yes	Yes	Buyer
Refund initiated / completed	Yes	Yes	Yes	Buyer
KYC approved / rejected	Yes	Yes	Yes	Seller
Product approved / rejected	Yes	—	Yes	Seller
Settlement released	Yes	Yes	Yes	Seller
Low stock	Yes	—	Yes	Seller
Login from new device	—	Yes	Yes	User
Promotional campaign	Opt-in	Opt-in	Opt-in	Buyer

M19.3 Functional requirements

ID	Requirement	Pri
FR-M19-01	Templates are database records with a channel, an event key, variable placeholders, and — for SMS — the DLT template identifier.	P0
FR-M19-02	Admin can edit template content without a code deployment.	P0
FR-M19-03	All SMS is sent through SMS India Hub using DLT-registered sender IDs and pre-approved templates.	P0
FR-M19-04	Push notifications are delivered by FCM to registered device tokens. FCM is used for nothing else.	P0
FR-M19-05	Email is sent through a queued SMTP sender with retry — never synchronously inside a request.	P0

ID	Requirement	Pri
FR-M19-06	Every notification attempt is recorded with channel, template, recipient, status and provider reference.	P0
FR-M19-07	Failed sends retry with backoff; permanent failures alert operations.	P0
FR-M19-08	Transactional and promotional messages are separated; promotional requires opt-in and honours opt-out.	P0
FR-M19-09	Users manage notification preferences per channel for promotional messages only.	P0
FR-M19-10	Device tokens are registered on login and invalidated on logout or token refresh failure.	P0
FR-M19-11	An in-app notification centre shows recent notifications with read state.	P1
FR-M19-12	Admin can send a broadcast to a filtered user segment, subject to opt-in.	P1

DLT timeline risk. TRAI DLT entity registration, sender ID approval and per-template registration take 3–10 working days and are client-owned. Templates cannot be changed freely after approval, so the final wording of every transactional SMS must be settled early. Staging runs against a sandbox harness so login remains testable while approvals are pending — but production launch cannot precede approval.

MODULE 20 · PHASE 1C · DAYS 40–62

Invoicing and Tax

GST invoices per sub-order · seller of record · HSN · TCS · TDS · credit notes

Blocked on decision D2 (\$44). The seller-of-record position determines who issues the invoice, whose GSTIN appears on it, and how TCS and TDS are treated. Invoice templates and the settlement arithmetic cannot be finalised until the client answers. This is on the critical path for Phase 1C.

M20.1 Functional requirements

ID	Requirement	Pri
FR-M20-01	A GST-compliant tax invoice is generated per sub-order, recording the seller of record explicitly.	P0
FR-M20-02	Invoices carry invoice number, date, both parties' details, GSTIN where applicable, HSN per line, taxable value, CGST/SGST/IGST split, and total.	P0
FR-M20-03	Tax split follows place of supply — intra-state produces CGST+SGST, inter-state produces IGST.	P0
FR-M20-04	Invoice numbering is sequential, gapless and unique per seller of record per financial year.	P0
FR-M20-05	Invoices are downloadable as PDF by buyer, seller and Admin.	P0
FR-M20-06	Financial values on an invoice are snapshotted; a later change to product GST rate never alters an issued invoice.	P0
FR-M20-07	Credit notes are issued on refund and cancellation, referencing the original invoice.	P0
FR-M20-08	TCS under GST s.52 is computed, deducted before settlement, and reported per period.	P0
FR-M20-09	TDS under IT s.194-O is computed, deducted, and reported per period per vendor.	P0
FR-M20-10	TCS and TDS appear as itemised lines on every settlement statement.	P0
FR-M20-11	Own Stock sales carry no TCS or TDS — the platform is the seller, not the facilitator.	P0

ID	Requirement	Pri
FR-M20-12	Period reports for GSTR-8 and TDS filing are exportable.	P0
FR-M20-13	B2B invoices carry the buyer's GSTIN captured before checkout.	P0
FR-M20-14	Packing slips are distinct from invoices and carry no pricing where configured.	P1

MODULE 21 · PHASE 1E

Reports and Analytics

Sales · settlement · inventory · vendor performance · own-stock separation

M21.1 Report catalogue

Report	Audience	Contents
Sales summary	Admin	Revenue, order count, AOV by period, with model, category, seller and buyer-type breakdowns.
Own Stock performance	Admin	Own-stock sales, margin and inventory movement, tracked separately from vendor stock .
Vendor performance	Admin	Order volume, acceptance rate, dispatch SLA adherence, cancellation rate, return rate, rating.
Commission earned	Admin	Commission by period, seller, category and product.
Settlement report	Admin, Seller	Gross, deductions, net, transfer references, status — by cycle.
Pending settlement	Admin, Seller	Held transfers with expected release dates and blocking reasons.
Inventory report	Admin, Seller	On hand, reserved, available, low-stock, by owner bucket.
Order fulfilment	Admin	Orders by state, ageing, stalled orders, exception queue.
Returns and RTO	Admin	Volume and reasons by seller, category and product; cost allocation.
Tax reports	Admin	GST output, TCS and TDS by period, ready for filing.
Catalog health	Admin	Products by approval state, missing mandatory attributes, zero-result searches.
Seller dashboard	Seller	Today's orders, pending dispatch, settlement due, low stock, rating.
Buyer order history	Buyer	Past orders, invoices, returns, wallet history where M16 is in scope.

M21.2 Functional requirements

ID	Requirement	Pri
FR-M21-01	All reports support date-range filtering and are scoped to the requester's permissions.	P0
FR-M21-02	Own-stock figures are separable from vendor figures in every financial and inventory report.	P0
FR-M21-03	Reports are exportable to CSV, gated behind the <code>export</code> permission.	P0
FR-M21-04	Financial reports reconcile to the settlement ledger to the rupee.	P0
FR-M21-05	Sellers see only their own data in every report.	P0

ID	Requirement	Pri
FR-M21-06	Dashboards render within the p95 latency target at full data volume, using pre-aggregation where needed.	P0
FR-M21-07	Large exports run asynchronously with a completion notification.	P1
FR-M21-08	Scheduled report delivery by email.	P2

MODULE 22 · PHASE 1E

CMS and Merchandising

Banners · homepage layout · static pages · policy versioning · SEO

M22.1 Functional requirements

ID	Requirement	Pri
FR-M22-01	Admin manages homepage banners with image, link target, position, and a scheduled active window.	P0
FR-M22-02	Admin curates homepage sections — featured categories, featured products, deals — with ordering.	P0
FR-M22-03	Static pages (About, Contact, FAQ, policies) are editable through a rich-text editor.	P0
FR-M22-04	Policy documents are versioned. Publishing a new version records it and preserves prior versions, so an acceptance record always resolves to the exact text accepted.	P0
FR-M22-05	A material policy change can require re-acceptance from users on next login.	P0
FR-M22-06	SEO metadata is editable for products, categories and static pages.	P1
FR-M22-07	Banners are separately configurable for web and app surfaces.	P1
FR-M22-08	Grievance Officer details are maintained in the CMS and rendered where the Consumer Protection Rules require.	P0

MODULE 23 · PHASE 1G · DAYS 75–90

Audit, Security and Backup

Audit trail · 2FA · encryption · rate limiting · backups · restore drill

M23.1 Functional requirements

ID	Requirement	Pri
FR-M23-01	Every privileged action writes an audit entry: actor, entity, before state, after state, timestamp and IP.	P0
FR-M23-02	Audit logs are append-only; no application path updates or deletes them.	P0
FR-M23-03	Audit logs are searchable by actor, entity, action type and date range.	P0
FR-M23-04	Every KYC document access is logged with actor and document.	P0
FR-M23-05	Two-factor authentication is available and enforceable by policy for Admin, Staff and Seller.	P0
FR-M23-06	KYC documents and PII are encrypted at rest.	P0
FR-M23-07	Rate limiting applies across auth, OTP, order placement and export endpoints, with distinct thresholds.	P0

ID	Requirement	Pri
FR-M23-08	Daily automated database backup plus off-site cloud backup.	P0
FR-M23-09	Restore is tested during hardening and documented as a runbook with a measured recovery time.	P0
FR-M23-10	Login alerts on new device or unusual location.	P1
FR-M23-11	Anti-fraud checks on order placement: velocity, address anomaly, prior chargeback, suspicious patterns.	P0
FR-M23-12	Secrets live in environment configuration; test and live keys never coexist in one environment.	P0
FR-M23-13	Data-subject rights — access, correction, erasure — are executable by Admin per the Privacy Policy.	P0
FR-M23-14	A breach-response runbook exists covering containment, assessment, and notification to the Data Protection Board, affected users and CERT-In.	P0

PART D

Surfaces

What each user actually sees. Three surfaces, four app binaries, one responsive web platform — all React and React Native, sharing components wherever possible.

37 Buyer app and website

38 Seller panel and app

39 Admin panel

37. Buyer app and website

Deliverables: Buyer Android app, Buyer iOS app, responsive website.

37.1 Screen inventory

Screen	Contents and behaviour
Splash / onboarding	First-run introduction, permission prompts for notifications.
Registration	Name, mobile, email; OTP verification; policy acceptance checkboxes with links to the live policy versions.
Login	Mobile + OTP, or email + password. Forgot password. Resend with cooldown timer.
Home	Banners, featured categories, featured products, deals, recently viewed. CMS-curated.
Search	Search bar with suggestions, recent searches, zero-result handling.
Category browse	Category tree navigation, subcategory tiles, product grid.
Product listing	Grid or list, faceted filters, sort control, infinite scroll or pagination, tier-correct prices.
Product detail	Image gallery, title, brand, tier price with MOQ where applicable, variant selector, stock state, delivery estimate by PIN, seller identity, specifications, reviews, related products, add to cart, add to wishlist.
Cart	Lines grouped by seller, quantity controls, MOQ messages, unavailable-line flags, price-change notices, coupon entry, totals breakdown.
Address book	List, add, edit, delete, set default. Used at checkout.
Checkout	Address selection, per-sub-order delivery estimates, GSTIN entry for B2B, coupon, wallet redemption where M16 is in scope, order summary, pay.
Payment	Gateway-hosted. Success, failure and retry states handled without losing the cart.
Order confirmation	Order reference, per-sub-order breakdown, expected delivery.
Order list	All orders with status chips, filterable by status and date.
Order detail	Per-sub-order status, tracking timeline with AWB, invoice download, cancel button where eligible, raise-return entry point, reorder.
Tracking	Courier, AWB, event history, current status, expected delivery.
Return request	Reason selection limited to the three accepted reasons, mandatory photo upload, description, submission confirmation.
Wishlist	Saved products, move to cart, remove.
Reviews	Write review with rating, text and photos, from a delivered order.
Wallet ADD	Balance, ledger history with reasons, expiry notices. Only if M16 is commissioned.
Profile	Personal details, change mobile or email with verification, password, 2FA, notification preferences, sessions.
Notifications	In-app notification centre with read state.
Static pages	About, Contact, FAQ, Privacy Policy, T&C, Shipping Policy, Return Policy, Grievance Officer details.

37.2 B2B-specific behaviour

- Tier prices replace retail prices throughout once the buyer's role is approved.
- MOQ is displayed on the product detail page and enforced in the cart with an explicit message.
- GSTIN capture is required before checkout completes.
- Reorder is prominent, since B2B buyers repeat similar orders.
- Bulk quantity entry is supported without repeated tapping of a stepper.

38. Seller panel and app

Deliverables: Seller Android app, Seller iOS app, web Seller Panel with responsive dashboard.

38.1 Screen inventory

Screen	Contents and behaviour
Registration	Store name, business model, business type, contact. Leads into KYC.
KYC upload	Per-document upload with type labels, status per document, reviewer notes on rejection, resubmission.
Onboarding status	Krozenda approval state <i>and</i> payment-partner verification state, shown separately and honestly.
Dashboard	Today's orders, pending dispatch count, settlement due, low-stock alerts, rating, recent activity.
Products list	Own catalog with approval-state chips, search, filters, bulk actions.
Product create / edit	Type selection, all mandatory attributes, variant matrix builder, images, tier pricing, MOQ, category and brand selection.
Bulk import	Template download, file upload, validation preview, error report download, commit.
Inventory	On hand, reserved, available per SKU; individual and bulk update; low-stock threshold configuration.
Pricing	Tier prices by buyer role, MOQ per tier, discount rules within platform limits.
Orders	Assigned sub-orders with accept and reject actions, filters by status and urgency, SLA countdown.
Order detail	Items, buyer delivery details, packing slip print, weight and dimension entry, AWB generation, label download.
Shipments	AWB list, pickup status, tracking, label reprint.
Returns	Claims against this seller, evidence photos, response to Admin, resolution outcome.
Settlements	Released settlements with full arithmetic, pending settlements with expected release date and blocking reason, deduction detail, downloadable statement.
Commission	Applicable rates, effective dates, notice of upcoming changes.
Reports	Sales, fulfilment SLA, returns, inventory — scoped to this seller.
Business documents	Stored KYC documents, expiry tracking, re-upload on change.
Store settings	Store profile, pickup and warehouse addresses, bank details (triggering re-verification), settlement cycle preference.
Profile and security	Contact details, password, 2FA, sessions, notification preferences.

Design priority for the Seller Panel. Sellers care most about two questions: "what must I ship today" and "when am I paid and how much". The dashboard and the settlements screen should answer both without a support ticket. Every settlement figure must be traceable to its arithmetic on screen, not only in an export.

39. Admin panel

Deliverable: React web Admin Panel. Not a mobile app.

39.1 Feature matrix

Area	Capabilities
Dashboard	Revenue, orders, new sellers, pending approvals, exception queues, stalled orders, settlement due today.
Users	Directory across all thirteen roles; search, filter, view, assign roles and capabilities, suspend, reinstate, force reset, export.
Roles & permissions	Create, edit and clone permission sets; assign to roles; view the effective matrix. Super Admin only.
Sellers	Application queue, per-document review, approve or reject with reason, request more documents, linked-account state, suspend, performance metrics.
Companies, partners, dealers, traders, distributors	Role-filtered views over the same seller and buyer management surface.
Categories	Tree management, approval, licence requirements, default GST and commission, activation.
Brands	Registry, approval queue, brand requests from sellers.
Products	All products across all sellers and own stock; approval queue with bulk actions; edit; delist; import oversight.
Inventory	Own-stock management with adjustment reasons; visibility over vendor stock; low-stock overview.
Orders	All orders and sub-orders; filters across every dimension; force-cancel with reason; exception queue; stalled-order report.
Payments	Payment records, reconciliation exceptions, orphan and duplicate payments, refund initiation.
Settlements	Settlement ledger, pending and released, manual hold and release, reversal tracking, recoverables, cycle configuration per vendor.
Payouts ADD	Payout runs, maker-checker approval, per-beneficiary status, run reports. Only if M15.5 is commissioned.
Commission	Rule management across all five scopes, simulation tool, change history.
Returns	Claim queue with evidence, vendor input, approve or reject with reason, resolution and cost allocation.
Coupons & offers	Create, scope, schedule, usage caps, cost-bearer configuration, redemption reporting.
Banners & CMS	Homepage curation, banners with scheduling, static pages, policy versioning and re-acceptance triggers.
Reports & analytics	The full catalogue in §M21.1 with export.
Notifications	Template management, delivery logs, failure investigation, segment broadcast.

Area	Capabilities
Audit logs	Searchable append-only trail across all privileged actions.
Backup	Backup status, last successful run, restore runbook access. Super Admin only.
Settings	Platform configuration — return window, SLA thresholds, serviceability, tax rates, grievance officer details.

PART E

Delivery

How the ninety days are structured, how completion is judged, what is tested, what is handed over — and the decisions and risks that determine whether the timeline holds.

40	Phase plan and timeline
41	Acceptance criteria
42	Testing strategy
43	Deployment and handover
44	Open decisions register
45	Risk register
46	Client-owned dependencies
47	Glossary

40. Phase plan and timeline

40.1 Phase 0 — Foundation, days 1–10

Blocking for everything else.

- Repository structure, monorepo decision, branching strategy, CI.
- Environment setup — dev, staging, production — and secrets management.
- MongoDB schema for the core collections in §9.
- Auth service: registration, login, JWT and refresh, email verification, and our own OTP implementation delivered via SMS India Hub.
- RBAC engine — permissions as data, thirteen roles seeded.
- Design system and shared component library for React and React Native.
- Razorpay sandbox plus a Route linked-account test flow.
- Shiprocket sandbox connection.
- Firebase project and FCM wiring.
- SMS India Hub and SMTP sandbox harness.
- API contract definition (OpenAPI) so frontend teams work against it from day one.

Exit criteria. A user can register, verify, log in, and hit a permission-gated endpoint on staging. Nothing in Phase 1 starts until this is demonstrated.

40.2 Phase 1 — Core platform, days 11–90

Block	Days	Scope	Modules
1A	11–30	Catalog and seller onboarding	M03, M04, M05, M06, M08
1B	25–45	Pricing, cart and checkout	M07, M09, M10, M11, M17
1C	40–62	Order and fulfilment	M12, M13, M20
1D	55–72	Settlement, returns and notifications	M14, M15, M19
1E	30–80	Panels and apps (parallel)	Part D surfaces, M18, M21, M22
1F	60–75	B2B core	Tiered pricing, MOQ, bulk quantity, tax invoice
1G	75–90	Hardening and handover	M23, testing, UAT, deployment, training

40.3 Detailed block contents

1A — Catalog and seller onboarding (days 11–30)

Seller and company registration with KYC upload; Admin manual verification with approve/reject and reason; digital policy acceptance capture; category tree and brand management with approval chains; product CRUD across all five types; SKU, barcode, HSN, GST and MOQ fields; CSV and Excel import with validation and error reporting; product approval workflow; search infrastructure per decision D3.

1B — Pricing, cart and checkout (days 25–45)

Commission engine with the full resolution hierarchy; multiple price levels by buyer role; discount and pricing rules; cart with per-line tier resolution; multi-vendor cart splitting into parent order and sub-orders; address book with multiple addresses; coupons and offers; Razorpay checkout with webhook verification and signature validation.

1C — Order and fulfilment (days 40–62)

Full order lifecycle state machine; auto supplier and vendor assignment with order forwarding; vendor accept and reject; packing slip generation; GST-compliant invoice per sub-order; Shiprocket AWB, label, courier allocation and pickup scheduling; tracking sync by webhook; delivery confirmation; cancellation rules for buyer and admin pre-dispatch; RTO handling.

1D — Settlement, returns and notifications (days 55–72)

Razorpay Route transfers, holds and reversals; settlement scheduling across automatic, manual, weekly and monthly cycles; settlement ledger and reports; returns and replacement workflow; refund and partial refund; notification engine across FCM, SMS India Hub with DLT templates, and SMTP; template management as configurable records.

1E — Panels and apps (days 30–80, parallel)

All four app binaries are React Native builds; web surfaces are React. Buyer, Seller and Admin surfaces per Part D.

1F — B2B core (days 60–75)

Dealer, distributor, trader and company registration; MOQ; bulk quantity; wholesale pricing; dealer pricing; multiple price levels; tax invoice.

1G — Hardening and handover (days 75–90)

Security hardening, 2FA, rate limiting, encryption at rest; performance optimisation against the §12 targets; functional, performance and security testing; UAT cycle with the client; production deployment with domain and hosting configuration; daily and cloud backup configuration; administrator training; technical documentation, API documentation and deployment guide; handover of source code, git repository, database, server credentials and all configs.

Timeline integrity. The 90 working days cover Phase 1 only. The two scope additions in §5.4 — wallet (M16) and automated payouts (M15.5) — are **not** inside this commitment. If the client commissions either, the timeline and cost must be renegotiated rather than absorbed.

41. Acceptance criteria

41.1 Definition of done — per requirement

A functional requirement is complete when all of the following hold:

- The behaviour is implemented and demonstrable on staging.
- Validation and error paths behave as specified, not only the happy path.
- Permission gating is enforced and verified for every role that should and should not have access.
- Audit logging is present where the requirement involves a privileged action.
- The relevant surface reflects it — an API without its UI is not done.
- It is covered by at least one test at the appropriate level.

41.2 End-to-end acceptance scenarios

ID	Scenario	Pass condition
E2E-01	Buyer registration to first order	Register, verify by OTP, browse, add to cart, checkout, pay in sandbox, receive confirmation, see the order.
E2E-02	Seller onboarding to live listing	Register, upload KYC, accept policies, Admin approves, linked account created, product created and approved, product visible on storefront.

ID	Scenario	Pass condition
E2E-03	Multi-vendor order split	Cart with items from two sellers plus own stock produces three sub-orders, three invoices, three independent statuses.
E2E-04	Partial cancellation	Cancelling one sub-order refunds proportionally, reverses only that transfer, releases only that stock, and leaves siblings untouched.
E2E-05	Full fulfilment cycle	Vendor accepts, packs, generates AWB, ships; tracking updates arrive; delivery confirmed; sub-order reaches Delivered.
E2E-06	Settlement release	Held transfer releases automatically after delivery plus return window; settlement statement arithmetic matches the order to the rupee.
E2E-07	Return with evidence	Buyer raises a Damaged claim with photos, Admin approves, refund issued, transfer reversed, both legs reconciled.
E2E-08	RTO cycle	Undelivered shipment returns to origin; transfer reversed, inventory restored, buyer refunded per attribution rule.
E2E-09	B2B tiered purchase	Dealer sees dealer pricing, MOQ enforced, GSTIN captured, tax invoice issued with correct GSTIN.
E2E-10	Bulk import	1,000-row CSV imports valid rows, rejects invalid rows, and produces an error report identifying row, column and reason.
E2E-11	RBAC enforcement	Each role can reach exactly its permitted endpoints and no others; a seller cannot read another seller's records.
E2E-12	Reconciliation	A deliberately dropped webhook is detected by the nightly job and corrected, with an exception reported.

41.3 Non-functional acceptance

- Load test at 1,500 concurrent users sustains the p95 < 500 ms target.
- Catalog of 100,000 SKUs returns search and listing pages within target.
- 5,000 orders in a day processed including the settlement job, without backlog.
- Backup runs daily and a restore is demonstrated with a measured recovery time.
- Security testing finds no critical or high-severity issues outstanding at handover.

42. Testing strategy

Level	Coverage	Notes
Unit	Commission resolution, price tier resolution, settlement arithmetic, state-machine transitions, wallet ledger where in scope	The financial logic is where defects are most expensive; it is tested exhaustively including boundary cases.
Integration	Module services against a test database; provider interfaces against mocks	Because integrations sit behind interfaces, the whole system is testable without live third-party calls.
Contract	API responses against the OpenAPI specification	Prevents frontend and backend drifting apart.
End-to-end	The twelve scenarios in §41.2	Run against staging with sandbox payment and logistics.

Level	Coverage	Notes
Performance	The §12.1 targets	Requires client confirmation of the targets first.
Security	Auth bypass, privilege escalation, injection, XSS, CSRF, IDOR, rate-limit evasion, webhook forgery	Includes deliberate attempts to release settlement without delivery and to forge webhook signatures.
UAT	Client-run against real business scenarios	Days 75–90. Requires client availability; a UAT participant list should be named by day 60.

42.1 Financial testing — special attention

Money paths get adversarial testing rather than confirmatory testing. Specifically: attempts to release a held transfer without a delivery event; refunds issued without a matching transfer reversal; concurrent redemption or checkout on the last unit; duplicate webhook delivery; replayed webhook payloads; payout submission retried after a timeout; and reversal where the vendor balance is insufficient. Each must fail safely and leave a reconcilable record.

43. Deployment and handover

43.1 Environments

Environment	Purpose and configuration
Development	Local and shared dev. All integrations in sandbox. Seeded reference data.
Staging	Production-shaped. Sandbox payment and logistics. UAT runs here. Sandbox SMS harness while DLT approval is pending.
Production	Live keys, live DLT templates, live domains, backups enabled, monitoring active.

43.2 Handover deliverables

Deliverable	Contents
Source code and git repository	Full history, branching documentation, transferred ownership.
Database	Schema, seed scripts, current production data.
Server credentials and configuration	All environment configuration, transferred securely and rotated at handover.
Technical documentation	Architecture, module map, data model, integration notes.
API documentation	OpenAPI specification with examples.
Deployment guide	Build, deploy, rollback, environment variables, health checks.
Runbooks	Backup and restore, breach response, reconciliation exception handling, settlement hold and release.
App store artefacts	Four signed binaries, store listings, signing keys transferred to client accounts.
Administrator training	Live sessions covering seller approval, product approval, order exceptions, returns adjudication, settlement operations, reporting.

Post-delivery support is not included. Support, warranty period and AMC are explicit exclusions and would be a separate engagement. The handover package is designed so the client's own team can operate and extend the platform without ongoing dependency.

44. Open decisions register

These determine the critical path. Each needs a named owner and a date. Items D1, D2 and D3 are blocking.

ID	Decision	Blocks	Consequence of delay
D1	Is Razorpay Route provisioned on the client's account?	M15, M03.09, Phase 1D	Route is not enabled by default and requires Razorpay's approval. No vendor settlement can be built or tested without it. This is the single most important item in this document — every day of delay is a day off the settlement block, which sits at days 55–72 with no slack behind it.
D2	Seller of record per business model	M20, M15, T&C	Determines who invoices the buyer, whose GSTIN appears, who carries consumer-law liability, and the TCS/TDS treatment. Invoice templates and settlement arithmetic cannot be finalised without it. Blocks Phase 1C.
D3	Search: Atlas Search or Elasticsearch?	M08, M05	MongoDB regex will not scale to 100,000 SKUs with facets. This determines how products are <i>indexed</i> , not merely queried, so deciding late forces a rewrite of the catalog read path. Must be settled before catalog work begins at day 11.
D4	Transfer on hold at capture, or transfer on delivery?	M15	Appzeto recommends on-hold at capture (M15.3). Affects what vendors see and when, and how cleanly reversals work.
D5	No-Return policy versus Consumer Protection (E-Commerce) Rules	M14, T&C	The blanket No-Return default must be reconciled with consumer law by the client's counsel, per category. The build implements what the client decides; it will not choose on the client's behalf.
D6	Penny-drop bank verification — in or out?	M03, M15	Currently excluded. Without it, bank-detail typos surface as failed transfers rather than being caught at entry. See risk R-04.
D7	Who bears the payment gateway fee?	M15, Vendor Agreement	Platform-absorbed or deducted from the vendor. Materially affects vendor economics and must be stated in the Vendor Agreement.
D8	Wallet (M16) — commission it or not?	M16, M10, M15	Scope addition. Requires a change request for cost and timeline, and legal confirmation that the closed-loop design avoids PPI authorisation. Also requires a funded platform float, since wallet redemption must not short-pay vendors (M16.5).
D9	Automated payouts (M15.5) — commission it or not?	M15.5	Scope addition requiring a RazorpayX account. Confirm that disbursement-only is intended and that statutory payroll computation is handled elsewhere.
D10	Non-functional targets confirmed?	\$12, \$42	The \$12.1 figures are Appzeto's proposals. Without client confirmation, performance testing has no pass criteria.
D11	Which two supplier integrations?	M05.13	Phase 1 is capped at two. The specific suppliers and their API documentation must be identified early; each is bespoke adapter work.
D12	Return window length	M14, M15	Feeds directly into the settlement release gate. A longer window delays every vendor payout.

45. Risk register

ID	Risk	Impact	Mitigation
R-01	Route not provisioned in time	High	Raise with Razorpay in week one. Build the settlement module against the provider interface with a mock so logic is complete and tested before live credentials arrive; only the adapter then needs wiring.
R-02	DLT approval delays block OTP in production	High	Start registration in week one. Staging uses a sandbox harness so login stays testable. Templates are data, so approved text swaps in without a deploy.
R-03	Seller-of-record decision arrives late	High	Model the invoice entity so the seller of record is a stored field rather than derived logic, limiting rework to templates and tax configuration.
R-04	No penny-drop verification — vendor bank typos	Med	Manual comparison of cancelled cheque against entered account number, recorded as a review checklist item. Razorpay's own linked-account KYC catches much but not all. Escalate to D6.
R-05	Dropshipping stock drift causing unfulfillable orders	Med	Vendor rejection is a first-class state with a reliable cancel-and-refund path. Rejection rates are reported per supplier so the commercial problem is visible.
R-06	Search decision deferred past day 11	High	Escalate D3 immediately. Catalog indexing design depends on it; late change means rewriting the read path.
R-07	Refund issued without transfer reversal	High	The two legs are one business operation with nightly reconciliation and a morning exception report. Adversarially tested per §42.1.
R-08	Duplicate payout on retry	High	Deterministic idempotency keys derived from period and beneficiary, never random. Called out as the highest-risk detail in the payout build.
R-09	Webhook loss causing silent state drift	Med	Nightly reconciliation against the provider's records converts an unbounded class of silent errors into a bounded report.
R-10	Scope creep from the two ADD items absorbing Phase 1 time	High	Both are marked ADD throughout and excluded from the 90-day commitment. Neither starts without a signed change request.
R-11	Client UAT availability compresses hardening	Med	Name UAT participants by day 60; run UAT in parallel with hardening rather than after it.
R-12	Legal policy text not supplied	Med	Policy acceptance flow is built against versioned CMS records, so final text drops in without code change. Cannot go live without it.
R-13	App store review delays	Med	Submit builds for review before the end of hardening; account setup is a client dependency that must complete by day 60.
R-14	Wallet short-paying vendors if M16 is rushed	High	M16.5 states the requirement explicitly: the platform funds the wallet portion so the vendor is made whole. Tested as a financial scenario.

46. Client-owned dependencies — tracking

Dependency	Needed by	Owner	Status
Razorpay account with Route enabled	Day 10	[Name]	[]
Razorpay MDR and transaction charges agreed	Day 10	[Name]	[]

Dependency	Needed by	Owner	Status
RazorpayX account (only if M15.5 commissioned)	On CR	[Name]	[]
Shiprocket account + per-vendor pickup locations	Day 35	[Name]	[]
TRAI DLT registration, Sender ID, template approval	Day 20	[Name]	[]
SMTP credentials + SPF / DKIM / DMARC records	Day 20	[Name]	[]
Firebase project ownership	Day 10	[Name]	[]
Domain and hosting	Day 70	[Name]	[]
Apple Developer + Google Play Console accounts	Day 60	[Name]	[]
Legal policy text — Privacy, T&C, Shipping, Return, Vendor Agreement	Day 60	[Name]	[]
Seller-of-record and GST TCS/TDS position (D2)	Day 30	[Name]	[]
Logo, brand assets, product content	Day 25	[Name]	[]
Supplier API documentation for the two integrations (D11)	Day 25	[Name]	[]
Commission structure and settlement cycle policy	Day 25	[Name]	[]
UAT participant list	Day 60	[Name]	[]
UI/UX design and design system	Day 15	Appzeto	In progress

47. Glossary

Term	Meaning
AWB	Air Waybill — the courier's tracking number for a shipment.
B2B / B2C	Business-to-business / business-to-consumer buyer types.
Capability	A named ability held by a user, checked by the authorisation guard. Distinct from a role.
Commission snapshot	The commission figure written onto a sub-order at placement and never recomputed.
DLT	Distributed Ledger Technology registry mandated by TRAI for commercial SMS in India.
DPDP Act	Digital Personal Data Protection Act, 2023.
FCM	Firebase Cloud Messaging — used for push notifications only.
HSN	Harmonised System of Nomenclature — the tax classification code for a product.
Linked account	A vendor's account within Razorpay Route, holding its own KYC and receiving its share of split payments.
MOQ	Minimum Order Quantity applicable to a B2B price tier.

Term	Meaning
Own Stock	Model C — inventory owned by the platform and fulfilled from its own warehouse.
PA framework	RBI's Payment Aggregator regulations governing who may collect and settle funds on behalf of merchants.
PPI	Prepaid Payment Instrument — a regulated category that a reloadable wallet would fall into.
Route	Razorpay's split-settlement product, mandatory for vendor settlement here.
RTO	Return to Origin — a shipment returned to the seller because it could not be delivered.
Sub-order	The per-seller portion of an order; the unit of fulfilment, invoicing, refund and settlement.
TCS	Tax Collected at Source under GST s.52, applicable to marketplace operators.
TDS	Tax Deducted at Source under Income-tax s.194-O, applicable to e-commerce operators.
Tier	A price level tied to a buyer role — retail, wholesale, dealer or distributor.

Document status. This PRD specifies the contracted Phase 1 delivery in full. Two modules — wallet (M16) and automated payouts (M15.5) — are marked **ADD** and sit outside the 90-day commitment pending a change request. Twelve decisions in §44 remain open; D1, D2 and D3 are blocking and should be resolved before Phase 0 exits. Where this document and the signed Scope of Work conflict, the Scope of Work prevails.

Krozenda Product Requirements Document · Version 1.0 · [DD Month YYYY] · Prepared by Appzeto for [Legal Entity Name Pvt. Ltd.] · Quote ref #0001710